# A First Course in Abstract Algebra

## Part 2: Permutations, Cosets, and Direct Products

Detailed worked solutions

Mathjock

**Text and numbering.** John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

**How to read the solutions.** Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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# 8 Groups of Permutations

**Conventions.** Permutations compose from right to left. The identity is $e$. A group generated by an element $a$ is denoted $\langle a \rangle$.

### Exercise 8.1: Product $\tau\sigma$

**Plan.** Apply the rightmost permutation first. The intermediate images are

| $x$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $\sigma(x)$ | 3 | 1 | 4 | 5 | 6 | 2 |
| $\tau(\sigma(x))$ | 1 | 2 | 3 | 6 | 5 | 4 |

The final row is the image of $1, \ldots, 6$ under the required product. Thus

$$\tau\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 2 & 3 & 6 & 5 & 4 \end{pmatrix}.$$

### Exercise 8.2: Product $\tau^2\sigma$

**Plan.** Apply the rightmost permutation first. The intermediate images are

| $x$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $\sigma(x)$ | 3 | 1 | 4 | 5 | 6 | 2 |
| $\tau(\sigma(x))$ | 1 | 2 | 3 | 6 | 5 | 4 |
| $\tau(\tau(\sigma(x)))$ | 2 | 4 | 1 | 5 | 6 | 3 |

The final row is the image of $1, \ldots, 6$ under the required product. Thus

$$\tau^2\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 4 & 1 & 5 & 6 & 3 \end{pmatrix}.$$

### Exercise 8.3: Product $\mu\sigma^2$

**Plan.** Apply the rightmost permutation first. The intermediate images are

| $x$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $\sigma(x)$ | 3 | 1 | 4 | 5 | 6 | 2 |
| $\sigma(\sigma(x))$ | 4 | 3 | 5 | 6 | 2 | 1 |
| $\mu(\sigma(\sigma(x)))$ | 3 | 4 | 1 | 6 | 2 | 5 |

The final row is the image of $1, \ldots, 6$ under the required product. Thus

$$\mu\sigma^2 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 4 & 1 & 6 & 2 & 5 \end{pmatrix}.$$

### Exercise 8.4: Product $\sigma^{-2}\tau$

**Plan.** Apply the rightmost permutation first. The intermediate images are

| $x$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $\tau(x)$ | 2 | 4 | 1 | 3 | 6 | 5 |
| $\sigma^{-1}(\tau(x))$ | 6 | 3 | 2 | 1 | 5 | 4 |
| $\sigma^{-1}(\sigma^{-1}(\tau(x)))$ | 5 | 1 | 6 | 2 | 4 | 3 |

The final row is the image of $1, \ldots, 6$ under the required product. Thus

$$\sigma^{-2}\tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 5 & 1 & 6 & 2 & 4 & 3 \end{pmatrix}.$$

### Exercise 8.5: Product $\sigma^{-1}\tau\sigma$

**Plan.** Apply the rightmost permutation first. The intermediate images are

| $x$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $\sigma(x)$ | 3 | 1 | 4 | 5 | 6 | 2 |
| $\tau(\sigma(x))$ | 1 | 2 | 3 | 6 | 5 | 4 |
| $\sigma^{-1}(\tau(\sigma(x)))$ | 2 | 6 | 1 | 5 | 4 | 3 |

The final row is the image of $1, \ldots, 6$ under the required product. Thus

$$\sigma^{-1}\tau\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 6 & 1 & 5 & 4 & 3 \end{pmatrix}.$$

### Exercise 8.6: Order of the subgroup generated by $\sigma$

Following 1 gives $1 \mapsto 3 \mapsto 4 \mapsto 5 \mapsto 6 \mapsto 2 \mapsto 1$. Thus $\sigma = (1, 3, 4, 5, 6, 2)$. It takes exactly six applications to return 1, and six applications fix every point. Therefore $\sigma$ has order 6, so $|\langle\sigma\rangle| = 6$.

### Exercise 8.7: Order of the subgroup generated by $au^2$

The cycles of $\tau$ are $(1, 2, 4, 3)(5, 6)$. Squaring advances two steps in each cycle, giving $\tau^2 = (1, 4)(2, 3)$, with $5, 6$ fixed. This permutation is not the identity, but its square is. Consequently $\langle\tau^2\rangle = \{e, \tau^2\}$ and the requested order is 2.

### Exercise 8.8: The power $\sigma^{100}$

Since $\sigma^6 = e$ and $100 = 16 \cdot 6 + 4$, $\sigma^{100} = \sigma^4$. Advance four steps around $(1, 3, 4, 5, 6, 2)$ to obtain

$$\sigma^{100} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 5 & 2 & 1 & 3 & 4 \end{pmatrix} = (1, 6, 4)(2, 5, 3).$$

For example $1 \mapsto 3 \mapsto 4 \mapsto 5 \mapsto 6$, checking the first image.

### Exercise 8.9: The power $\mu^{100}$

Here $\mu = (1,5)(3,4)$, so $\mu^2 = e$. Since 100 is even, $\mu^{100} = (\mu^2)^{50} = e$. The points $2, 6$ are fixed throughout.

### Exercise 8.10: Partition into isomorphism classes

**Plan.** First separate finite groups by order, then use commutativity, torsion, and divisibility to distinguish infinite groups. The complete partition is

$$\{(\mathbb{Z},+),(17\mathbb{Z},+),(3\mathbb{Z},+),\langle\pi\rangle\},$$
$$\{\mathbb{Z}_6, G\},\quad \{\mathbb{Z}_2, S_2\},\quad \{S_6\},$$
$$\{(\mathbb{Q},+)\},\quad \{(\mathbb{R},+),(\mathbb{R}^+,\cdot)\},$$
$$\{(\mathbb{R}^*,\cdot)\},\quad \{(\mathbb{Q}^*,\cdot)\},\quad \{(\mathbb{C}^*,\cdot)\}.$$

The maps $n \mapsto 17n$, $n \mapsto 3n$, and $n \mapsto \pi^n$ give the first class. The generator specified for $G$ is $(1,3,4)(2,5)$, of order 6, so $G \cong \mathbb{Z}_6$. Both $S_2$ and $\mathbb{Z}_2$ are cyclic of order 2. The group $S_6$ has 720 elements and is nonabelian. The exponential $x \mapsto e^x$ identifies $(\mathbb{R},+)$ with $(\mathbb{R}^+,\cdot)$. The additive group $\mathbb{Q}$ is countable and divisible (every equation $nx = q$, $n > 0$, has a solution), whereas $\mathbb{Z}$ is not divisible. The group $\mathbb{Q}^*$ is countable but has the element $-1$ of order 2, distinguishing it from the countable torsion-free groups. The uncountable group $\mathbb{R}^*$ has exactly two torsion elements, $1, -1$, and is not divisible because $x^2 = -1$ has no real solution. The group $\mathbb{C}^*$ has roots of unity of every positive order. These observations also exclude all remaining possible identifications.

### Exercise 8.11: Orbit under $\sigma$

Repeatedly applying the permutation gives $1 \mapsto 3 \mapsto 4 \mapsto 5 \mapsto 6 \mapsto 2 \mapsto 1$. Negative powers traverse the same cycle backwards, so the orbit of 1 is exactly $\{1,2,3,4,5,6\}$.

### Exercise 8.12: Orbit under $au$

Repeatedly applying the permutation gives $1 \mapsto 2 \mapsto 4 \mapsto 3 \mapsto 1$. Negative powers traverse the same cycle backwards, so the orbit of 1 is exactly $\{1,2,3,4\}$.

### Exercise 8.13: Orbit under $\mu$

Repeatedly applying the permutation gives $1 \mapsto 5 \mapsto 1$. Negative powers traverse the same cycle backwards, so the orbit of 1 is exactly $\{1,5\}$.

### Exercise 8.14: Six geometric labels for $S_3$

Let $r$ be a $120^\circ$ rotation of the equilateral triangle and $s$ its specified reflection. The rotations are $e, r, r^2$. Applying $s$ first and then each rotation gives the three reflections $s, rs, r^2s$, whose reflection axes pass through the three different vertices. They are distinct, and no reflection is a rotation because reflections reverse orientation. Hence $e, r, r^2, s, rs, r^2s$ are precisely the six elements of $S_3$.

### Exercise 8.15: Eight geometric labels for $D_4$

Choose the quarter-turn $r = \rho_1$ and any one reflection $s$, for example $s = \mu_1$. The eight elements are

$$e, r, r^2, r^3, s, rs, r^2s, r^3s.$$

The first four preserve orientation and give the four rotational placements of a square. The last four reverse orientation and send a fixed vertex to four distinct vertices, so they are all different reflections. Thus they give every element of $D_4$.

### Exercise 8.16: Permutations fixing 3

Fixing 3 in a permutation of $\{1, 2, 3, 4\}$ leaves a free permutation of $\{1, 2, 4\}$. There are $3 \cdot 2 \cdot 1 = 6$ such choices, and each extends uniquely by $3 \mapsto 3$. The answer is 6.

### Exercise 8.17: Permutations sending 2 to 5

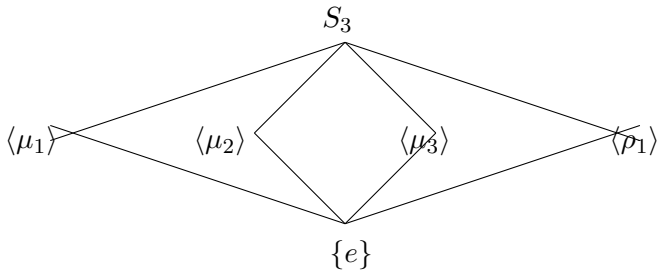
The image of 2 is already fixed. The other four arguments must map bijectively to the four unused targets $1, 2, 3, 4$. There are $4! = 24$ ways to do this. Notice that fixing an image different from the argument does not change the count.

### Exercise 8.18: Subgroups of $S_3$

**(a)** Both nonidentity rotations generate $\{e, \rho_1, \rho_2\}$, since $\rho_1^2 = \rho_2$ and $\rho_1^3 = e$. Since $\mu_1^2 = e$, $\langle \mu_1 \rangle = \{e, \mu_1\}$. **(b)** The entire subgroup list is

$$\{e\}, \quad \langle \mu_1 \rangle, \langle \mu_2 \rangle, \langle \mu_3 \rangle, \quad \langle \rho_1 \rangle, \quad S_3.$$

An element is either a reflection or a rotation. Two distinct reflections have a nontrivial rotation as product, and a reflection together with a nontrivial rotation generates all six elements. Thus there are no additional subgroups. In the subgroup diagram $\{e\}$ lies below each of the four proper nontrivial subgroups, and each of those lies below $S_3$; there are no inclusions between those four.




### Exercise 8.19: All subgroups of $D_4$

Write $r = \rho_1$ and $s = \mu_1$, so $r^4 = s^2 = e$ and $sr = r^{-1}s$. The cyclic subgroups generated by single elements are

$$\{e\}, \quad \langle r^2 \rangle, \quad \langle r \rangle, \quad \langle s \rangle, \langle rs \rangle, \langle r^2 s \rangle, \langle r^3 s \rangle.$$

Here $\langle r\rangle$ has order 4, while the five other nontrivial cyclic subgroups have order 2. If a subgroup contains a reflection and $r$ or $r^3$, it is all of $D_4$. If it contains two reflections $r^i s, r^j s$, their product is $r^{i-j}$. For odd $i-j$ this again generates $D_4$; for even nonzero $i-j$, it produces one of the two Klein groups

$$K_0 = \{e, r^2, s, r^2 s\}, \qquad K_1 = \{e, r^2, rs, r^3 s\}.$$

These are all possibilities for subgroups with two or more generators. Thus the diagram has top $D_4$; the next level $\langle r\rangle, K_0, K_1$; the next level the five order-2 groups; and bottom $\{e\}$. The group $\langle r^2\rangle$ connects to all three groups of order 4, $\langle s\rangle, \langle r^2 s\rangle$ connect to $K_0$, and $\langle rs\rangle, \langle r^3 s\rangle$ connect to $K_1$. These are exactly the edges in the supplied diagram, verifying it exhaustively.

### Exercise 8.20: The cyclic subgroup generated by $ho$

The permutation is $\rho = (1,2,4)(3,5)$, so its order is $\operatorname{lcm}(3,2) = 6$. Indeed, $\rho^k = e$ precisely when both $3 \mid k$ and $2 \mid k$. In the following table $e = \rho^0 = \rho^6$:

| | $e$ | $\rho$ | $\rho^2$ | $\rho^3$ | $\rho^4$ | $\rho^5$ |
|---|---|---|---|---|---|---|
| $e$ | $e$ | $\rho^1$ | $\rho^2$ | $\rho^3$ | $\rho^4$ | $\rho^5$ |
| $\rho^1$ | $\rho^1$ | $\rho^2$ | $\rho^3$ | $\rho^4$ | $\rho^5$ | $e$ |
| $\rho^2$ | $\rho^2$ | $\rho^3$ | $\rho^4$ | $\rho^5$ | $e$ | $\rho^1$ |
| $\rho^3$ | $\rho^3$ | $\rho^4$ | $\rho^5$ | $e$ | $\rho^1$ | $\rho^2$ |
| $\rho^4$ | $\rho^4$ | $\rho^5$ | $e$ | $\rho^1$ | $\rho^2$ | $\rho^3$ |
| $\rho^5$ | $\rho^5$ | $e$ | $\rho^1$ | $\rho^2$ | $\rho^3$ | $\rho^4$ |

The rule is $\rho^i \rho^j = \rho^{i+j \bmod 6}$. This group is abelian and cyclic, so it is not isomorphic to the nonabelian group $S_3$, despite having the same order.

### Exercise 8.21: Six permutation matrices

**(a)** Each displayed matrix has exactly one 1 in each row and column, and all other entries zero. Such a matrix permutes the standard basis vectors. If $P_\sigma e_j = e_{\sigma(j)}$, then

$$P_\sigma P_\tau e_j = P_\sigma e_{\tau(j)} = e_{\sigma(\tau(j))},$$

so $P_\sigma P_\tau = P_{\sigma\tau}$. Thus the six matrices are closed under multiplication, contain $I$, and have inverses $P_{\sigma^{-1}}$; associativity comes from matrix multiplication. **(b)** There are exactly $3! = 6$ permutations of three basis vectors, and these are the six displayed matrices. The map $\sigma \mapsto P_\sigma$ is a bijective homomorphism, so the group is isomorphic to $S_3$.

### Exercise 8.22: Matrices representing $D_4$

Use the rotation and reflection matrices

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad S = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The requested eight matrices are $I, R, R^2, R^3, S, RS, R^2S, R^3S$, explicitly

$$\begin{pmatrix}1&0\\0&1\end{pmatrix}, \begin{pmatrix}0&-1\\1&0\end{pmatrix}, \begin{pmatrix}-1&0\\0&-1\end{pmatrix}, \begin{pmatrix}0&1\\-1&0\end{pmatrix},$$
$$\begin{pmatrix}1&0\\0&-1\end{pmatrix}, \begin{pmatrix}0&1\\1&0\end{pmatrix}, \begin{pmatrix}-1&0\\0&1\end{pmatrix}, \begin{pmatrix}0&-1\\-1&0\end{pmatrix}.$$

They act on the square with vertices $(\pm1, \pm1)$ as exactly its eight symmetries. The equations $R^4 = S^2 = I$ and $SR = R^{-1}S$ prove closure and the same multiplication rule as $D_4$, giving the desired isomorphism.

### Exercise 8.23: Symmetries of the three-armed figure

The three equal arms meet at 120°. Every symmetry fixes their common junction and permutes the three endpoints. Rotations through $0°, 120°, 240°$ and the three reflections through an arm realize all six permutations of those endpoints. The action is faithful because an isometry fixing three noncollinear endpoints is the identity. Thus the group is $D_3 \cong S_3$.

### Exercise 8.24: Symmetries of the divided circle

The distinguished diameter must map to itself. The possibilities are the identity, the half-turn, reflection in that diameter, and reflection in its perpendicular diameter. Each nonidentity element squares to the identity, and the product of the two reflections is the half-turn. Hence the symmetry group is the Klein four-group $V \cong \mathbb{Z}_2 \times \mathbb{Z}_2$.

### Exercise 8.25: Symmetries of the cross

The four equal arms have endpoints forming a square. A symmetry of the cross gives a symmetry of those endpoints, and every symmetry of the square preserves the cross. The four rotations and four reflections therefore give exactly $D_4$.

### Exercise 8.26: Symmetries of the infinite decorated line

The downward vertical teeth and upward slanting teeth distinguish the two sides of the horizontal line and also distinguish left from right. Thus neither a reflection nor a half-turn preserves the figure. The allowed translations are exactly integral multiples of the repeating horizontal spacing $a$. Their composition adds the integers: $T_{ma}T_{na} = T_{(m+n)a}$. Hence the symmetry group is the infinite cyclic group $\mathbb{Z}$.

### Exercise 8.27: Regular representations

**Left regular representation of $\mathbb{Z}_4$.** For $k \in \mathbb{Z}_4$, put $L_k(x) = k + x \pmod 4$. The four permutations are

$$L_0 = e, \quad L_1 = (0,1,2,3), \quad L_2 = (0,2)(1,3), \quad L_3 = (0,3,2,1),$$

with table

| | $L_0$ | $L_1$ | $L_2$ | $L_3$ |
|---|---|---|---|---|
| $L_0$ | $L_0$ | $L_1$ | $L_2$ | $L_3$ |
| $L_1$ | $L_1$ | $L_2$ | $L_3$ | $L_0$ |
| $L_2$ | $L_2$ | $L_3$ | $L_0$ | $L_1$ |
| $L_3$ | $L_3$ | $L_0$ | $L_1$ | $L_2$ |

**Right regular representation of $S_3$.** The book's homomorphic convention is $R_a(x) = xa^{-1}$, not $xa$. With columns in the order $\rho_0, \rho_1, \rho_2, \mu_1, \mu_2, \mu_3$, the images are

| | $\rho_0$ | $\rho_1$ | $\rho_2$ | $\mu_1$ | $\mu_2$ | $\mu_3$ |
|---|---|---|---|---|---|---|
| $R_{\rho_0}$ | $\rho_0$ | $\rho_1$ | $\rho_2$ | $\mu_1$ | $\mu_2$ | $\mu_3$ |
| $R_{\rho_1}$ | $\rho_2$ | $\rho_0$ | $\rho_1$ | $\mu_3$ | $\mu_1$ | $\mu_2$ |
| $R_{\rho_2}$ | $\rho_1$ | $\rho_2$ | $\rho_0$ | $\mu_2$ | $\mu_3$ | $\mu_1$ |
| $R_{\mu_1}$ | $\mu_1$ | $\mu_3$ | $\mu_2$ | $\rho_0$ | $\rho_2$ | $\rho_1$ |
| $R_{\mu_2}$ | $\mu_2$ | $\mu_1$ | $\mu_3$ | $\rho_1$ | $\rho_0$ | $\rho_2$ |
| $R_{\mu_3}$ | $\mu_3$ | $\mu_2$ | $\mu_1$ | $\rho_2$ | $\rho_1$ | $\rho_0$ |

For example $R_{\rho_1}(\rho_0) = \rho_2$. Since $R_a R_b(x) = (xb^{-1})a^{-1} = x(ab)^{-1}$, the multiplication table of the representation is

| | $R_{\rho_0}$ | $R_{\rho_1}$ | $R_{\rho_2}$ | $R_{\mu_1}$ | $R_{\mu_2}$ | $R_{\mu_3}$ |
|---|---|---|---|---|---|---|
| $R_{\rho_0}$ | $R_{\rho_0}$ | $R_{\rho_1}$ | $R_{\rho_2}$ | $R_{\mu_1}$ | $R_{\mu_2}$ | $R_{\mu_3}$ |
| $R_{\rho_1}$ | $R_{\rho_1}$ | $R_{\rho_2}$ | $R_{\rho_0}$ | $R_{\mu_3}$ | $R_{\mu_1}$ | $R_{\mu_2}$ |
| $R_{\rho_2}$ | $R_{\rho_2}$ | $R_{\rho_0}$ | $R_{\rho_1}$ | $R_{\mu_2}$ | $R_{\mu_3}$ | $R_{\mu_1}$ |
| $R_{\mu_1}$ | $R_{\mu_1}$ | $R_{\mu_2}$ | $R_{\mu_3}$ | $R_{\rho_0}$ | $R_{\rho_1}$ | $R_{\rho_2}$ |
| $R_{\mu_2}$ | $R_{\mu_2}$ | $R_{\mu_3}$ | $R_{\mu_1}$ | $R_{\rho_2}$ | $R_{\rho_0}$ | $R_{\rho_1}$ |
| $R_{\mu_3}$ | $R_{\mu_3}$ | $R_{\mu_1}$ | $R_{\mu_2}$ | $R_{\rho_1}$ | $R_{\rho_2}$ | $R_{\rho_0}$ |

In each case evaluation at the identity distinguishes the permutations, so the maps from the original group are injective.

### Exercise 8.28: Definition of a permutation

A permutation of $S$ is a *bijection* $f : S \to S$: it must be both one-to-one and onto. The original wording omits surjectivity. For instance $n \mapsto n + 1$ on $\{0, 1, 2, \ldots\}$ is one-to-one but misses 0, and therefore is not a permutation.

### Exercise 8.29: Definition of the left regular representation

The definition is correct: $g \mapsto L_g$, with $L_g(x) = gx$, maps $G$ into $S_G$. To check the terms are justified, $L_g^{-1} = L_{g^{-1}}$, so each $L_g$ is a permutation, and $L_g L_h = L_{gh}$. Finally $L_g(e) = g$, so this representation is one-to-one.

### Exercise 8.30: Translation as a permutation

Yes. If $x + 1 = y + 1$, subtracting 1 gives $x = y$, so the map is one-to-one. Given a real target $u$, choosing $x = u - 1$ gives $f(x) = u$, so it is onto. Its inverse is $u \mapsto u - 1$.

### Exercise 8.31: Squaring as a permutation

No. The two inputs 1 and $-1$ have the same image 1, so the map is not one-to-one. It also misses every negative real number because $x^2 \geq 0$. Either failure excludes it from $S_{\mathbb{R}}$.

### Exercise 8.32: The map $x \mapsto -x^3$

Yes. If $-x^3 = -y^3$, then $x^3 = y^3$ and hence $x = y$, since the real cube function is strictly increasing. Given $u \in \mathbb{R}$, choose $x = \sqrt[3]{-u}$. Then $-x^3 = u$, so the map is onto. Its inverse is $u \mapsto \sqrt[3]{-u}$.

### Exercise 8.33: The exponential map

No. Although $e^x$ is strictly increasing and therefore one-to-one, its range is $(0, \infty)$, not all of $\mathbb{R}$. In particular there is no real input with image 0. It would be a bijection from $\mathbb{R}$ to $\mathbb{R}^+$, but a permutation here must map $\mathbb{R}$ onto itself.

### Exercise 8.34: A cubic polynomial

The polynomial is $f(x) = x^3 - x^2 - 2x = x(x-2)(x+1)$. Thus $f(-1) = f(0) = f(2) = 0$ at three distinct inputs, so it is not one-to-one and is not a permutation. (It is onto $\mathbb{R}$ by continuity and its opposite infinite end limits, but that does not repair injectivity.)

### Exercise 8.35: True or false

(a) True: a permutation is a bijection and hence injective.

(b) False: injectivity does not ensure surjectivity or even equal domain and codomain; the shift on nonnegative integers is a counterexample.

(c) True: if a finite set has $n$ elements, a surjection onto all $n$ targets cannot repeat an image, or at most $n - 1$ targets would occur.

(d) True: Cayley's theorem uses $g \mapsto L_g$ to embed $G$ into $S_G$.

(e) True: if $ab = ba$ for all elements in a group, the same equality holds when both elements are restricted to any subgroup.

(f) True: $\langle a \rangle = \{a^k : k \in \mathbb{Z}\}$ is closed under products and inverses and contains $e$.

(g) False: $S_{10}$ has 10!, not 10, elements.

(h) False: $S_3$ is nonabelian, whereas every cyclic group is abelian.

(i) False: $S_1$ is trivial and cyclic, and $S_2$ is cyclic of order 2.

(j) True: this is the permutation-group form of Cayley's theorem.

### Exercise 8.36: A nonabelian group with abelian proper subgroups

Take $S_3$. Its proper subgroups are $\{e\}$, the three order-2 groups generated by transpositions, and the cyclic order-3 group $A_3$. All are cyclic and therefore abelian. Nevertheless $(1,2)(2,3) = (1,2,3)$ and $(2,3)(1,2) = (1,3,2)$ differ, so $S_3$ itself is nonabelian.

### Exercise 8.37: All self-maps under composition

Function composition of maps $A \to A$ is an associative binary operation and has identity $\mathrm{id}_A$. Thus the set of all self-maps is a monoid (and in particular a semigroup). It is generally not a group: if $A$ has at least two elements, a constant map is not injective and cannot have an inverse. If $A$ is a singleton, there is only the identity map and it is the trivial group.

### Exercise 8.38: A Cayley digraph for $D_n$

Let $r$ be a rotation of order $n$ and $s$ a reflection, so $sr = r^{-1}s$. Label the two rows (or two concentric $n$-gons) by $r^i$ and $r^i s$, with $i$ read modulo $n$. For right-multiplication arcs, draw

$$r^i \xrightarrow{\ r\ } r^{i+1}, \qquad r^i s \xrightarrow{\ r\ } r^{i-1}s, \qquad r^i \text{—}_s\, r^i s.$$

Thus the two $n$-cycles have opposite directions and a reflection edge joins corresponding vertices. This fully specifies the digraph, with $2n$ vertices and the correct outgoing arc of each type at every vertex. Schematically:

$$\begin{array}{ccccccccccc} e & \longrightarrow & r & \longrightarrow & r^2 & \longrightarrow \cdots \longrightarrow & r^{n-1} & \longrightarrow & e \\ \updownarrow s & & \updownarrow s & & \updownarrow s & & \updownarrow s & & \\ s & \longleftarrow & rs & \longleftarrow & r^2 s & \longleftarrow \cdots \longleftarrow & r^{n-1}s & \longleftarrow & s \end{array}$$

The reflection arcs are undirected because $s^2 = e$.

### Exercise 8.39: Synopsis of Cayley's theorem

Each $g \in G$ determines a permutation $L_g : x \mapsto gx$ of $G$, and $L_g L_h = L_{gh}$, so $g \mapsto L_g$ is a homomorphism into $S_G$. It is injective because $L_g(e) = g$, hence identifies $G$ with the subgroup $\{L_g : g \in G\}$.

### Exercise 8.40: The stabilizer of one point

Yes. The identity fixes $b$. If $\sigma(b) = b$ and $\tau(b) = b$, then $(\sigma\tau)(b) = \sigma(b) = b$. Also $\sigma(b) = b$ implies $\sigma^{-1}(b) = b$. Thus the set contains the identity and is closed under composition and inverses, and is a subgroup of $S_A$.

### Exercise 8.41: Permutations sending a point into a subset

Not necessarily. Let $A = \{1,2,3\}$, $B = \{1,2\}$, and $b = 1$. The permutations $\sigma = (2,3)$ and $\tau = (1,2)$ both send 1 into $B$, but $(\sigma\tau)(1) = \sigma(2) = 3 \notin B$. Thus the proposed set can fail closure under products.

### Exercise 8.42: Permutations mapping a subset into itself

Not necessarily, because $B$ may be infinite. Let $A = \mathbb{Z}$ and $B = \{0, 1, 2, \ldots\}$. The shift $\sigma(n) = n + 1$ satisfies $\sigma[B] \subseteq B$, but its inverse sends 0 to $-1 \notin B$. Hence the set can fail closure under inverses. If $B$ is finite, inclusion does force equality, but no finiteness is assumed here.

### Exercise 8.43: The setwise stabilizer of a subset

Yes. The identity maps $B$ onto itself. If $\sigma[B] = B$ and $\tau[B] = B$, then $(\sigma\tau)[B] = \sigma[\tau[B]] = \sigma[B] = B$. Applying $\sigma^{-1}$ to $\sigma[B] = B$ gives $\sigma^{-1}[B] = B$. These three facts prove that the set is a subgroup.

### Exercise 8.44: Order and rotation subgroup of $D_n$

Choose the image of one vertex: $n$ possibilities. Its adjacent vertex must go to one of the two adjacent vertices of the chosen image: 2 possibilities. These choices determine the entire polygon placement, and all occur through rotations or reflections. Therefore $|D_n| = 2n$. The rotations through multiples of $2\pi/n$ form the cyclic subgroup $\langle r \rangle = \{e, r, \ldots, r^{n-1}\}$ of order $n$, exactly half of $2n$.

### Exercise 8.45: Rotations of a cube

Choose which of the six faces is on the bottom and then which of its four neighboring faces points forward. These choices determine a cube rotation, so the rotation group has $6 \cdot 4 = 24$ elements. For each of the three axes through centers of opposite faces, the quarter-turns form a distinct cyclic subgroup $\{e, r, r^2, r^3\}$ of order 4. For each of the four long diagonals joining opposite vertices, the $120^\circ$ rotations form a distinct cyclic subgroup of order 3. Different axes give different nonidentity rotations, establishing the requested lower bounds on numbers of subgroups.

### Exercise 8.46: Noncommutativity of $S_n$

For $n \geq 3$, let $a = (1, 2)$ and $b = (2, 3)$, fixing all other symbols. Then $(ab)(1) = a(1) = 2$, whereas $(ba)(1) = b(2) = 3$. Thus $ab \neq ba$ and $S_n$ is nonabelian. Comparing the images of one point is enough to distinguish two permutations.

### Exercise 8.47: Only the identity commutes with all permutations

**Plan.** Show a permutation moving a point fails to commute with a suitable transposition. Suppose $\sigma(a) = b \neq a$. Choose $c$ different from $a, b$, possible because $n \geq 3$, and let $\tau = (b, c)$. Then $\tau(a) = a$, so $(\sigma\tau)(a) = b$, while $(\tau\sigma)(a) = \tau(b) = c$. These images differ. Hence no nonidentity $\sigma$ commutes with every permutation. The identity does commute with every permutation, proving the claim.

### Exercise 8.48: Intersecting orbits coincide

Suppose $c = \sigma^m(a) = \sigma^n(b)$ lies in both orbits. Applying $\sigma^{-n}$ gives $b = \sigma^{m-n}(a)$. Therefore any $\sigma^k(b) = \sigma^{k+m-n}(a)$ lies in the orbit of $a$, giving $O_{b,\sigma} \subseteq O_{a,\sigma}$. Interchanging $a, b$ gives the reverse inclusion, so the two orbits are equal.

### Exercise 8.49: A finite cyclic transitive permutation group

Write the nonempty finite set as $A = \{a_1, \ldots, a_m\}$. Define $\sigma(a_i) = a_{i+1}$ for $i < m$ and $\sigma(a_m) = a_1$. The cycle $\sigma$ has order $m$, so $H = \langle\sigma\rangle$ has $|H| = m = |A|$. Given $a_i, a_j$, the power $\sigma^{j-i}$ carries $a_i$ to $a_j$ (with exponents modulo $m$). Thus $H$ is transitive. For $m = 1$ the same construction gives the identity group.

### Exercise 8.50: Transitivity and a single orbit

If $\langle\sigma\rangle$ is transitive and $a \in A$, then for every $b \in A$ some power $\sigma^k$ sends $a$ to $b$. Hence $O_{a,\sigma} = A$. Conversely, if this orbit is all of $A$, write any $b = \sigma^r(a)$ and $c = \sigma^s(a)$. Then $\sigma^{s-r}(b) = c$, so $\langle\sigma\rangle$ is transitive. The asserted existence of $a$ presupposes $A \neq \varnothing$; for the empty set transitivity is vacuous and this existential formulation needs that qualification.

### Exercise 8.51: The opposite group

**(a)** Viewing the written equations through glass reverses left-to-right multiplication order: $a * b = c$ reads as $b *' a = c$. The associativity equation is reversed on both sides, so it is still associativity for the reversed operation. The identity and inverse equations are likewise preserved with the factors interchanged. **(b)** Define $x *' y = y * x$. Closure follows from closure of $*$. Compute

$$(x *' y) *' z = z * (y * x) = (z * y) * x = x *' (y *' z),$$

using associativity of $*$. The old identity satisfies $e *' x = x * e = x$ and $x *' e = e * x = x$. The old inverse satisfies $x *' x^{-1} = x^{-1} * x = e$ and $x^{-1} *' x = x * x^{-1} = e$. Thus the reversed operation gives a group, called the opposite group.

### Exercise 8.52: The group of right translations

For $a \in G$, define $\rho_a(x) = xa$. Each is a permutation with inverse $\rho_{a^{-1}}$. Composition reverses the subscripts:

$$\rho_a\rho_b(x) = (xb)a = x(ba), \qquad \rho_a\rho_b = \rho_{ba}.$$

Thus these permutations contain the identity, products, and inverses and form a group. The map $a \mapsto \rho_a$ need not be a homomorphism. Instead use $\Phi(a) = \rho_{a^{-1}}$; then $\Phi(a)\Phi(b) = \rho_{b^{-1}a^{-1}} = \rho_{(ab)^{-1}} = \Phi(ab)$. It is onto the set of right translations, and equality of two such translations, evaluated at $e$, gives $a^{-1} = b^{-1}$ and then $a = b$. Hence $\Phi$ is an isomorphism.

### Exercise 8.53: Permutation matrices and the Klein group

**(a)** List the elements of a finite group $G$ in some order and let $L_g$ be left multiplication by $g$. Define $P_g e_x = e_{gx}$ on the standard basis indexed by $G$. Each $P_g$ is a permutation matrix. On every basis vector,

$$P_g P_h e_x = e_{ghx} = P_{gh} e_x,$$

so $g \mapsto P_g$ is a homomorphism. If $P_g = P_h$, applying both sides to $e_e$ gives $g = h$, so it is an isomorphism onto its image. **(b)** Order $V$ as $e, a, b, c$, with $a^2 = b^2 = c^2 = e$ and $ab = c$. Then

$$P_e = I_4, \quad P_a = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix},$$

$$P_b = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \quad P_c = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

For example $P_a e_e = e_a$ and $P_a e_b = e_c$, matching multiplication by $a$. The construction in part (a) already verifies every product, including $P_a P_b = P_c$.

# 9 Orbits, Cycles, and the Alternating Groups

**Conventions.** Permutations compose from right to left. The identity is $e$. A group generated by an element $a$ is denoted $\langle a \rangle$.

### Exercise 9.1: Orbits of a finite permutation

**Plan.** Start at an unused symbol, repeatedly apply the permutation until it returns, and repeat with a symbol not yet used. The nontrivial cycles are $(1, 5, 2)(4, 6)$. Each fixed point forms a singleton orbit, so the complete orbit list is

$$\{1, 2, 5\}, \quad \{4, 6\}, \quad \{3\}.$$

Every symbol occurs in exactly one listed set; negative powers move backwards within the same set and produce no further points.

### Exercise 9.2: Orbits of a finite permutation

**Plan.** Start at an unused symbol, repeatedly apply the permutation until it returns, and repeat with a symbol not yet used. The nontrivial cycles are $(1, 5, 8, 7)(2, 6, 3)$. Each fixed point forms a singleton orbit, so the complete orbit list is

$$\{1, 5, 7, 8\}, \quad \{2, 3, 6\}, \quad \{4\}.$$

Every symbol occurs in exactly one listed set; negative powers move backwards within the same set and produce no further points.

### Exercise 9.3: Orbits of a finite permutation

**Plan.** Start at an unused symbol, repeatedly apply the permutation until it returns, and repeat with a symbol not yet used. The nontrivial cycles are $(1, 2, 3, 5, 4)(7, 8)$. Each fixed point forms a singleton orbit, so the complete orbit list is

$$\{1, 2, 3, 4, 5\}, \quad \{7, 8\}, \quad \{6\}.$$

Every symbol occurs in exactly one listed set; negative powers move backwards within the same set and produce no further points.

### Exercise 9.4: Orbits of the integer shift by 1

The $r$th power sends an integer $a$ to $a + 1r$, where $r$ ranges over $\mathbb{Z}$. Hence the orbit of $a$ is $a + 1\mathbb{Z}$, its residue class modulo 1. The distinct orbits are

$$\mathbb{Z}.$$

The sign of the shift only changes the direction in which these same points are traversed.

### Exercise 9.5: Orbits of the integer shift by 2

The $r$th power sends an integer $a$ to $a + 2r$, where $r$ ranges over $\mathbb{Z}$. Hence the orbit of $a$ is $a + 2\mathbb{Z}$, its residue class modulo 2. The distinct orbits are

$$0 + 2\mathbb{Z}, \quad 1 + 2\mathbb{Z}.$$

The sign of the shift only changes the direction in which these same points are traversed.

### Exercise 9.6: Orbits of the integer shift by $-3$

The $r$th power sends an integer $a$ to $a + -3r$, where $r$ ranges over $\mathbb{Z}$. Hence the orbit of $a$ is $a + 3\mathbb{Z}$, its residue class modulo 3. The distinct orbits are

$$0 + 3\mathbb{Z}, \quad 1 + 3\mathbb{Z}, \quad 2 + 3\mathbb{Z}.$$

The sign of the shift only changes the direction in which these same points are traversed.

### Exercise 9.7: A product of cycles

**Plan.** Read the factors from right to left, leaving a symbol unchanged whenever it is absent from a factor. The images of the eight arguments are

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 4 & 1 & 3 & 5 & 8 & 6 & 2 & 7 \end{pmatrix}.$$

Following each chain of these images yields the disjoint-cycle answer

$$(1, 4, 5)(7, 8)(2, 5, 7) = (1, 4, 5, 8, 7, 2).$$

Any symbol omitted from the final cycles is fixed. The two-row form provides a check of every image, not just of one cycle.

### Exercise 9.8: A product of cycles

**Plan.** Read the factors from right to left, leaving a symbol unchanged whenever it is absent from a factor. The images of the eight arguments are

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 7 & 2 & 8 & 5 & 4 & 1 & 6 \end{pmatrix}.$$

Following each chain of these images yields the disjoint-cycle answer

$$(1, 3, 2, 7)(4, 8, 6) = (1, 3, 2, 7)(4, 8, 6).$$

Any symbol omitted from the final cycles is fixed. The two-row form provides a check of every image, not just of one cycle.

## Exercise 9.9: A product of cycles

**Plan.** Read the factors from right to left, leaving a symbol unchanged whenever it is absent from a factor. The images of the eight arguments are

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 5 & 4 & 3 & 7 & 8 & 6 & 2 & 1 \end{pmatrix}.$$

Following each chain of these images yields the disjoint-cycle answer

$$(1,2)(4,7,8)(2,1)(7,2,8,1,5) = (1,5,8)(2,4,7).$$

Any symbol omitted from the final cycles is fixed. The two-row form provides a check of every image, not just of one cycle.

## Exercise 9.10: Disjoint cycles and transpositions

### 1. Follow the images.

The nontrivial chains are

$$1 \mapsto 8 \mapsto 1; \quad 3 \mapsto 6 \mapsto 4 \mapsto 3; \quad 5 \mapsto 7 \mapsto 5.$$

Thus the disjoint-cycle expression is $(1,8)(3,6,4)(5,7)$; any missing point is fixed.

### 2. Split each cycle.

Use $(a_1, a_2, \ldots, a_k) = (a_1, a_k) \cdots (a_1, a_3)(a_1, a_2)$, verified by following each $a_i$. This gives

$$(1,8)(3,6,4)(5,7) = (1,8)(3,4)(3,6)(5,7).$$

## Exercise 9.11: Disjoint cycles and transpositions

### 1. Follow the images.

The nontrivial chains are

$$1 \mapsto 3 \mapsto 4 \mapsto 1; \quad 2 \mapsto 6 \mapsto 2; \quad 5 \mapsto 8 \mapsto 7 \mapsto 5.$$

Thus the disjoint-cycle expression is $(1,3,4)(2,6)(5,8,7)$; any missing point is fixed.

### 2. Split each cycle.

Use $(a_1, a_2, \ldots, a_k) = (a_1, a_k) \cdots (a_1, a_3)(a_1, a_2)$, verified by following each $a_i$. This gives

$$(1,3,4)(2,6)(5,8,7) = (1,4)(1,3)(2,6)(5,7)(5,8).$$

## Exercise 9.12: Disjoint cycles and transpositions

**1. Follow the images.**

The nontrivial chains are

$$1 \mapsto 3 \mapsto 4 \mapsto 7 \mapsto 8 \mapsto 6 \mapsto 5 \mapsto 2 \mapsto 1.$$

Thus the disjoint-cycle expression is $(1, 3, 4, 7, 8, 6, 5, 2)$; any missing point is fixed.

**2. Split each cycle.**

Use $(a_1, a_2, \ldots, a_k) = (a_1, a_k) \cdots (a_1, a_3)(a_1, a_2)$, verified by following each $a_i$. This gives

$$(1, 3, 4, 7, 8, 6, 5, 2) = (1, 2)(1, 5)(1, 6)(1, 8)(1, 7)(1, 4)(1, 3).$$

## Exercise 9.13: Orders of cycles and permutations

(a) $(1, 4, 5, 7)$ has order 4: four applications return each listed point, and no smaller positive number returns 1.

(b) A cycle of length $k$ has order $k$. Each application advances one step, so a power fixes the cycle exactly when the exponent is divisible by $k$.

(c) The disjoint cycles of $\sigma = (4, 5)(2, 3, 7)$ have lengths $2, 3$, giving order 6. Those of $\tau = (1, 4)(3, 5, 7, 8)$ have lengths $2, 4$, giving order 4.

(d) Exercises 10, 11, 12 have nontrivial cycle lengths $(2, 3, 2)$, $(3, 2, 3)$, and $(8)$, respectively. Their orders are $6, 6, 8$.

(e) The order of a permutation is the least common multiple of its disjoint-cycle lengths: a power is the identity precisely when it is a multiple of every cycle length.

## Exercise 9.14: Maximum element order in $S_5$

**Plan.** Disjoint-cycle lengths are positive integers summing to 5, and the order is their least common multiple. Thus check partitions, rather than individual permutations. Grouping the partitions by their largest part $L$, the largest possible least common multiple in each group is

| $L$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| maximum | 1 | 2 | 6 | 4 | 5 |

For example, these values can be reproduced by listing the remaining nondecreasing parts, each at most $L$, with sum $5 - L$, and taking the least common multiple with $L$. The overall maximum is 6, attained by disjoint cycles of lengths $2, 3$. This gives both an example and an exhaustive upper bound.

## Exercise 9.15: Maximum element order in $S_6$

**Plan.** Disjoint-cycle lengths are positive integers summing to 6, and the order is their least common multiple. Thus check partitions, rather than individual permutations. Grouping the partitions by

their largest part $L$, the largest possible least common multiple in each group is

| $L$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| maximum | 1 | 2 | 6 | 4 | 5 | 6 |

For example, these values can be reproduced by listing the remaining nondecreasing parts, each at most $L$, with sum $6 - L$, and taking the least common multiple with $L$. The overall maximum is 6, attained by disjoint cycles of lengths $1, 2, 3$. This gives both an example and an exhaustive upper bound.

### Exercise 9.16: Maximum element order in $S_7$

**Plan.** Disjoint-cycle lengths are positive integers summing to 7, and the order is their least common multiple. Thus check partitions, rather than individual permutations. Grouping the partitions by their largest part $L$, the largest possible least common multiple in each group is

| $L$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|
| maximum | 1 | 2 | 6 | 12 | 10 | 6 | 7 |

For example, these values can be reproduced by listing the remaining nondecreasing parts, each at most $L$, with sum $7 - L$, and taking the least common multiple with $L$. The overall maximum is 12, attained by disjoint cycles of lengths $3, 4$. This gives both an example and an exhaustive upper bound.

### Exercise 9.17: Maximum element order in $S_{10}$

**Plan.** Disjoint-cycle lengths are positive integers summing to 10, and the order is their least common multiple. Thus check partitions, rather than individual permutations. Grouping the partitions by their largest part $L$, the largest possible least common multiple in each group is

| $L$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| maximum | 1 | 2 | 6 | 12 | 30 | 12 | 21 | 8 | 9 | 10 |

For example, these values can be reproduced by listing the remaining nondecreasing parts, each at most $L$, with sum $10 - L$, and taking the least common multiple with $L$. The overall maximum is 30, attained by disjoint cycles of lengths $2, 3, 5$. This gives both an example and an exhaustive upper bound.

### Exercise 9.18: Maximum element order in $S_{15}$

**Plan.** Disjoint-cycle lengths are positive integers summing to 15, and the order is their least common multiple. Thus check partitions, rather than individual permutations. Grouping the partitions by their largest part $L$, the largest possible least common multiple in each group is

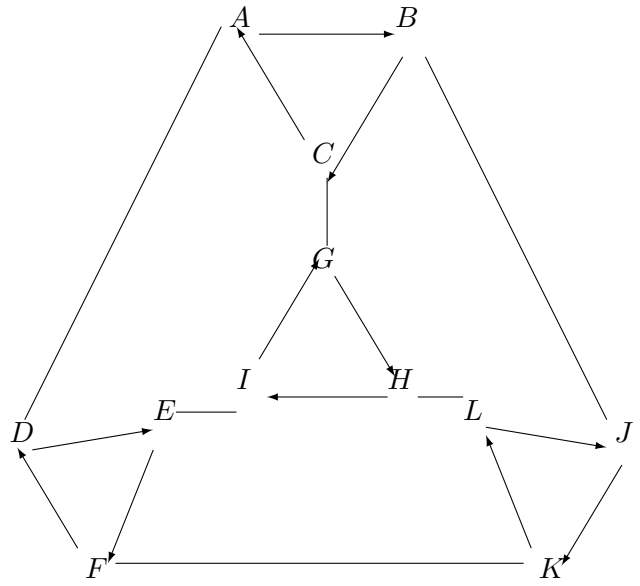
| $L$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| maximum | 1 | 2 | 6 | 12 | 60 | 60 | 105 | 56 | 45 | 30 | 44 | 12 | 26 | 14 | 15 |

For example, these values can be reproduced by listing the remaining nondecreasing parts, each at most $L$, with sum $15 - L$, and taking the least common multiple with $L$. The overall maximum is 105, attained by disjoint cycles of lengths $3, 5, 7$. This gives both an example and an exhaustive upper bound.

### Exercise 9.19: Labeling the Cayley digraph of $A_4$

Put $a = (1, 2, 3)$ and $b = (1, 2)(3, 4)$. Follow each solid arrow by right multiplying by $a$, and each dashed edge by right multiplying by $b$. The following table labels every position in the supplied diagram; the letters also appear in the schematic below.

| vertex | position | permutation |
|---|---|---|
| $A$ | $topleft$ | $e$ |
| $B$ | $topright$ | $(1, 2, 3)$ |
| $C$ | $belowtoppair$ | $(1, 3, 2)$ |
| $D$ | $leftoutside$ | $(1, 2)(3, 4)$ |
| $E$ | $leftinner$ | $(2, 4, 3)$ |
| $F$ | $bottomleft$ | $(1, 4, 3)$ |
| $G$ | $middletop$ | $(2, 3, 4)$ |
| $H$ | $middleright$ | $(1, 3)(2, 4)$ |
| $I$ | $middleleft$ | $(1, 4, 2)$ |
| $J$ | $rightoutside$ | $(1, 3, 4)$ |
| $K$ | $bottomright$ | $(1, 2, 4)$ |
| $L$ | $rightinner$ | $(1, 4)(2, 3)$ |




The six edges without arrowheads represent the dashed $b$ edges; both directions give the same operation since $b^2 = e$. For instance $D = b$, $E = ba$, and $F = ba^2$. The twelve entries are distinct and exhaust $A_4$, checking that no vertex has been missed.

### Exercise 9.20: Definition of an orbit

The proposed definition is correct when "minimal" means minimal under inclusion among nonempty invariant subsets: an orbit is a nonempty subset $O \subseteq A$ with $\sigma[O] = O$ and no proper nonempty

subset having that property. Indeed, $O_a = \{\sigma^k(a) : k \in \mathbb{Z}\}$ is invariant, and any nonempty invariant subset containing a point of $O_a$ contains its entire two-sided orbit. Conversely, pick $a$ in a minimal such set; its orbit lies inside the set and hence must equal it.

### Exercise 9.21: Definition of a cycle

A cycle in $S_n$ is a permutation with *at most one orbit having more than one element*. Its length is the size of its largest orbit. Fixed points outside the moving cycle produce additional singleton orbits, so "only one orbit" is too restrictive. The book includes the identity as a cycle of length 1.

### Exercise 9.22: Definition of the alternating group

Specify the common finite underlying set: the alternating group $A_n$ is the subgroup of $S_n$ consisting of all even permutations of $\{1, \ldots, n\}$. Without specifying this set or $n$, "all even permutations" does not describe a single group. Closure follows because products and inverses of even permutations are even.

### Exercise 9.23: True or false

(a) False: $(1, 2)(3, 4)$ has two nontrivial orbits.

(b) True: a cycle is a special kind of permutation.

(c) False: first one must prove that the parity of the number of transpositions is independent of the chosen expression; otherwise the definition could assign both parities to one permutation.

(d) False: $\langle(1, 2, 3, 4)\rangle$ has an odd permutation but its elements are $e$, a 4-cycle, a double transposition, and the inverse 4-cycle; none is a transposition.

(e) False: $|A_5| = 5!/2 = 60$.

(f) False: $S_2 = \{e, (1, 2)\}$ is cyclic.

(g) True: $A_3$ is cyclic of order 3.

(h) True: extend each permutation of $\{1, \ldots, 7\}$ by fixing 8.

(i) True: relabel the seven symbols other than 5, and extend by fixing 5; the particular omitted symbol is immaterial.

(j) False: the identity is even and so is absent, and the product of two odd permutations is even.

### Exercise 9.24: The table of $A_3$

The even permutations are $\rho_0 = e$, $\rho_1 = (1, 2, 3)$, and $\rho_2 = (1, 3, 2)$. Each 3-cycle is a product of two transpositions, while $\mu_1, \mu_2, \mu_3$ are single transpositions and are odd. Thus

| | $\rho_0$ | $\rho_1$ | $\rho_2$ |
|---|---|---|---|
| $\rho_0$ | $\rho_0$ | $\rho_1$ | $\rho_2$ |
| $\rho_1$ | $\rho_1$ | $\rho_2$ | $\rho_0$ |
| $\rho_2$ | $\rho_2$ | $\rho_0$ | $\rho_1$ |

The entries follow from $\rho_1^3 = e$ and $\rho_2 = \rho_1^2$.

### Exercise 9.25: Synopsis of the determinant proof

A permutation matrix obtained from the identity by $k$ row transpositions has determinant $(-1)^k$, so two transposition expressions for the same permutation must have the same parity because one matrix cannot have determinants 1 and $-1$ simultaneously.

### Exercise 9.26: Synopsis of the orbit-counting proof

Multiplying a permutation by one transposition merges two distinct orbits or splits one orbit in two, changing its number of orbits by exactly 1. Starting from the identity's $n$ orbits, the parity of the number of transpositions is therefore determined by the final number of orbits and cannot depend on the expression.

### Exercise 9.27: Bounds on numbers of transpositions

Let the disjoint-cycle lengths, including fixed points, be $k_1, \ldots, k_t$, with sum $n$. A $k_i$-cycle uses $k_i - 1$ transpositions, so the entire permutation uses

$$(k_1 - 1) + \cdots + (k_t - 1) = n - t.$$

**(a)** Since $t \geq 1$, this is at most $n - 1$. The identity can be represented by zero transpositions, the empty product. **(b)** A permutation which is not a cycle has at least two nontrivial orbits, hence $t \geq 2$, giving at most $n - 2$ transpositions. **(c)** If the permutation is odd, the displayed number $m = n - t$ is odd. Since $m \leq n - 1 < 2n + 3$ and $2n + 3 - m$ is even, insert $(2n + 3 - m)/2$ pairs $(1, 2)(1, 2)$ into that expression to obtain exactly $2n + 3$ factors. For an even permutation $m$ is even and the same insertion of $(2n + 8 - m)/2$ pairs gives exactly $2n + 8$ factors. Each pair is the identity and so does not change the permutation.

### Exercise 9.28: Merging orbits when a point is fixed

**(a)** Write the orbit containing $j$ as $(j, x_1, \ldots, x_k)$ and suppose $i$ is a different fixed point. Then direct calculation gives

$$(i, j)(j, x_1, \ldots, x_k) = (i, j, x_1, \ldots, x_k).$$

Thus the two old orbit diagrams and the new diagram can be drawn as

$$\underbrace{i \longrightarrow i}_{\text{one orbit}} \qquad \underbrace{j \longrightarrow x_1 \longrightarrow \cdots \longrightarrow x_k \longrightarrow j}_{\text{one orbit}} \quad \rightsquigarrow \quad i \longrightarrow j \longrightarrow x_1 \longrightarrow \cdots \longrightarrow x_k \longrightarrow i.$$

The other orbits are unchanged, so the orbit count decreases by one. **(b)** If $j$ is fixed too, the two singleton loops $i \to i$ and $j \to j$ become the single 2-cycle $i \to j \to i$ under $(i, j)\sigma$. Again exactly two orbits have become one.

### Exercise 9.29: Even permutations in a subgroup

If $H$ has no odd permutations, every element is even. Otherwise fix an odd $\tau \in H$. Left multiplication by $\tau$ maps the even elements of $H$ to odd elements of $H$, because signs multiply. Its inverse is left multiplication by $\tau^{-1}$, which is also odd. Thus it is a bijection between the even and odd elements of $H$. As $H$ is finite, exactly half its elements are even.

### Exercise 9.30: How many points a cycle moves

A cycle of length $n > 1$ moves exactly its $n$ listed points: each goes to the next distinct listed point, and all other points are fixed. Under the book's convention, a cycle of length 1 is the identity and moves 0 points. Thus the usual answer $n$ requires the nontrivial-cycle condition $n > 1$.

### Exercise 9.31: Finitely supported permutations form a subgroup

For a permutation $\sigma$, write $\operatorname{supp}(\sigma) = \{a : \sigma(a) \neq a\}$. The identity has empty support. If both $\sigma$ and $\tau$ fix a point $a$, their product fixes $a$, so

$$\operatorname{supp}(\sigma\tau) \subseteq \operatorname{supp}(\sigma) \cup \operatorname{supp}(\tau).$$

A union of two finite sets is finite, proving product closure. A point is fixed by $\sigma$ if and only if it is fixed by $\sigma^{-1}$, so the two supports are equal. Hence inverses also have finite support, and the given set is a subgroup.

### Exercise 9.32: A uniform bound on support size

No. Choose 100 distinct points of the infinite set $A$, and let $\sigma$ be a 50-cycle on the first 50, while $\tau$ is a 50-cycle on the other 50. Extend both by the identity elsewhere. Each moves exactly 50 points, so both belong to $K$. Their disjoint supports mean $\sigma\tau$ moves all 100 selected points, so it is not in $K$. Thus product closure fails.

### Exercise 9.33: The odd permutations form a coset

Let $\sigma$ be the fixed odd permutation and let $\pi$ be any odd permutation. Set $\alpha = \sigma^{-1}\pi$. Both factors are odd, so their product is even and $\alpha \in A_n$. Then $\pi = \sigma\alpha$. Conversely, the product of $\sigma$ with any even permutation is odd. Thus the set of odd permutations is exactly $\sigma A_n$.

### Exercise 9.34: Squaring an odd-length cycle

Let $\sigma = (a_0, a_1, \ldots, a_{n-1})$ with $n$ odd. Its square moves $a_i$ to $a_{i+2 \bmod n}$. If $2k \equiv 0 \pmod{n}$, oddness implies $n \mid k$, so the first positive return occurs after $n$ steps. Hence all $n$ symbols lie in one cycle of $\sigma^2$, and every point outside the original cycle stays fixed. For $n = 1$, both permutations are the identity, also a cycle by convention.

### Exercise 9.35: When a power of a cycle is a cycle

**Plan.** Advancing $r$ steps around an $n$-cycle partitions its $n$ points into $d = \gcd(n, r)$ cycles, each of length $n/d$. Indeed, $\sigma^{rk}(a) = a$ precisely when $n \mid rk$, whose least positive solution is $k = n/d$. If $n/d > 1$, all $d$ orbits are nontrivial, so the power is a cycle exactly when $d = 1$. There is an important convention in this text: the identity is also a cycle. If $n \mid r$, then $\sigma^r = e$ and it is therefore a cycle as well. With that convention the exact condition is

$$\boxed{\gcd(n, r) = 1 \quad \text{or} \quad n \mid r.}$$

If "cycle" is intended to mean a nonidentity $n$-cycle, the familiar condition is just $\gcd(n, r) = 1$ (for $n > 1$). The extra identity case must not be silently omitted under Definition 9.6.

### Exercise 9.36: Left multiplication is a permutation

The inverse of $\lambda_a : g \mapsto ag$ is $\lambda_{a^{-1}}$, because $a^{-1}(ag) = g$ and $a(a^{-1}g) = g$. Equivalently, cancellation gives injectivity, and a desired target $h$ is reached by the input $a^{-1}h$, giving surjectivity. Thus $\lambda_a \in S_G$.

### Exercise 9.37: The subgroup of left translations

By the preceding exercise every $\lambda_a$ is a permutation. The identity is $\lambda_e$. Associativity in $G$ gives

$$(\lambda_a \lambda_b)(g) = a(bg) = (ab)g = \lambda_{ab}(g),$$

so the set is closed under composition. Finally $\lambda_a^{-1} = \lambda_{a^{-1}}$ also lies in it. These facts prove it is a subgroup of $S_G$.

### Exercise 9.38: Left translations act transitively

Given $x, y \in G$, solve $ax = y$ by taking $a = yx^{-1}$. The left translation $\lambda_a$ then sends $x$ to $y$. Since this works for every pair of points in the set $G$, the left-translation subgroup is transitive on $G$.

### Exercise 9.39: Two generators for the symmetric group

Put $c = (1, 2, \ldots, n)$ and $t = (1, 2)$. Conjugating a transposition relabels its entries, so

$$c^k t c^{-k} = (c^k(1), c^k(2)).$$

For $k = 0, \ldots, n-2$, these include all adjacent transpositions $s_i = (i, i+1)$, $1 \le i < n$. For $i < j$ one has

$$(i, j) = s_i s_{i+1} \cdots s_{j-2} s_{j-1} s_{j-2} \cdots s_{i+1} s_i.$$

To verify it, the product before the middle transposition conjugates $(j-1, j)$ to $(i, j)$; alternatively follow $i, j$ and each intermediate point. Thus every transposition lies in $\langle c, t \rangle$. Every permutation is a product of transpositions, so $\langle c, t \rangle = S_n$.

# 10 Cosets and the Theorem of Lagrange

**Conventions.** Permutations compose from right to left. The identity is $e$. A group generated by an element $a$ is denoted $\langle a \rangle$.

### Exercise 10.1: Cosets of $4\mathbb{Z}$ in $\mathbb{Z}$

In additive notation, a coset is $a + 4\mathbb{Z} = \{a + 4k : k \in \mathbb{Z}\}$. Two representatives give the same coset precisely when their difference is divisible by 4. The distinct cosets in $\mathbb{Z}$ are

$$0 + 4\mathbb{Z}, \quad 1 + 4\mathbb{Z}, \quad 2 + 4\mathbb{Z}, \quad 3 + 4\mathbb{Z}.$$

The division algorithm shows every allowed representative belongs to one of these, and their different residues show the listed cosets are distinct.

### Exercise 10.2: Cosets of $4\mathbb{Z}$ in $2\mathbb{Z}$

In additive notation, a coset is $a + 4\mathbb{Z} = \{a + 4k : k \in \mathbb{Z}\}$. Two representatives give the same coset precisely when their difference is divisible by 4. The distinct cosets in $2\mathbb{Z}$ are

$$0 + 4\mathbb{Z}, \quad 2 + 4\mathbb{Z}.$$

The division algorithm shows every allowed representative belongs to one of these, and their different residues show the listed cosets are distinct.

### Exercise 10.3: Cosets of $\langle 2 \rangle$ in $\mathbb{Z}_{12}$

First list the subgroup generated by 2 by repeatedly adding it modulo 12:

$$H = \langle 2 \rangle = \{0, 2, 4, 6, 8, 10\}.$$

Adding a representative to every element of $H$ gives all distinct cosets:

$$0 + H = \{0, 2, 4, 6, 8, 10\},$$
$$1 + H = \{1, 3, 5, 7, 9, 11\}.$$

These sets are disjoint and together contain all 12 residues, verifying that the list is complete.

### Exercise 10.4: Cosets of $\langle 4 \rangle$ in $\mathbb{Z}_{12}$

First list the subgroup generated by 4 by repeatedly adding it modulo 12:

$$H = \langle 4 \rangle = \{0, 4, 8\}.$$

Adding a representative to every element of $H$ gives all distinct cosets:

$$0 + H = \{0, 4, 8\},$$
$$1 + H = \{1, 5, 9\},$$
$$2 + H = \{2, 6, 10\},$$
$$3 + H = \{3, 7, 11\}.$$

These sets are disjoint and together contain all 12 residues, verifying that the list is complete.

### Exercise 10.5: Cosets of $\langle 18\rangle$ in $\mathbb{Z}_{36}$

First list the subgroup generated by 18 by repeatedly adding it modulo 36:

$$H = \langle 18\rangle = \{0, 18\}.$$

Adding a representative to every element of $H$ gives all distinct cosets:

$$\begin{aligned}
0 + H &= \{0, 18\},\\
1 + H &= \{1, 19\},\\
2 + H &= \{2, 20\},\\
3 + H &= \{3, 21\},\\
4 + H &= \{4, 22\},\\
5 + H &= \{5, 23\},\\
6 + H &= \{6, 24\},\\
7 + H &= \{7, 25\},\\
8 + H &= \{8, 26\},\\
9 + H &= \{9, 27\},\\
10 + H &= \{10, 28\},\\
11 + H &= \{11, 29\},\\
12 + H &= \{12, 30\},\\
13 + H &= \{13, 31\},\\
14 + H &= \{14, 32\},\\
15 + H &= \{15, 33\},\\
16 + H &= \{16, 34\},\\
17 + H &= \{17, 35\}.
\end{aligned}$$

These sets are disjoint and together contain all 36 residues, verifying that the list is complete.

### Exercise 10.6: Left cosets of a reflection subgroup in $D_4$

Let $H = \{\rho_0, \mu_2\}$. For $a \in D_4$, the left coset is $aH = \{a, a\mu_2\}$. Reading the indicated products from Table 8.12 gives

$$\begin{aligned}
\rho_0 H &= \{\rho_0, \mu_2\},\\
\rho_1 H &= \{\rho_1, \delta_2\},\\
\rho_2 H &= \{\rho_2, \mu_1\},\\
\rho_3 H &= \{\rho_3, \delta_1\}.
\end{aligned}$$

The four pairs are disjoint and include all eight group elements, so these are all the left cosets.

### Exercise 10.7: Right cosets of the reflection subgroup

A right coset is $H\rho_i = \{\rho_i, \mu_2\rho_i\}$, so the list is

$$\begin{aligned} H\rho_0 &= \{\rho_0, \mu_2\}, \\ H\rho_1 &= \{\rho_1, \delta_1\}, \\ H\rho_2 &= \{\rho_2, \mu_1\}, \\ H\rho_3 &= \{\rho_3, \delta_2\}. \end{aligned}$$

These do not give the same partition as the left cosets: for example $\rho_1 H = \{\rho_1, \delta_2\}$ while $H\rho_1 = \{\rho_1, \delta_1\}$. Thus $H$ is not normal.

### Exercise 10.8: Reordered table for the reflection subgroup

Order the rows and columns in consecutive left cosets from Exercise 6. The resulting table is

| | $\rho_0$ | $\mu_2$ | $\rho_1$ | $\delta_2$ | $\rho_2$ | $\mu_1$ | $\rho_3$ | $\delta_1$ |
|---|---|---|---|---|---|---|---|---|
| $\rho_0$ | $\rho_0$ | $\mu_2$ | $\rho_1$ | $\delta_2$ | $\rho_2$ | $\mu_1$ | $\rho_3$ | $\delta_1$ |
| $\mu_2$ | $\mu_2$ | $\rho_0$ | $\delta_1$ | $\rho_3$ | $\mu_1$ | $\rho_2$ | $\delta_2$ | $\rho_1$ |
| $\rho_1$ | $\rho_1$ | $\delta_2$ | $\rho_2$ | $\mu_1$ | $\rho_3$ | $\delta_1$ | $\rho_0$ | $\mu_2$ |
| $\delta_2$ | $\delta_2$ | $\rho_1$ | $\mu_2$ | $\rho_0$ | $\delta_1$ | $\rho_3$ | $\mu_1$ | $\rho_2$ |
| $\rho_2$ | $\rho_2$ | $\mu_1$ | $\rho_3$ | $\delta_1$ | $\rho_0$ | $\mu_2$ | $\rho_1$ | $\delta_2$ |
| $\mu_1$ | $\mu_1$ | $\rho_2$ | $\delta_2$ | $\rho_1$ | $\mu_2$ | $\rho_0$ | $\delta_1$ | $\rho_3$ |
| $\rho_3$ | $\rho_3$ | $\delta_1$ | $\rho_0$ | $\mu_2$ | $\rho_1$ | $\delta_2$ | $\rho_2$ | $\mu_1$ |
| $\delta_1$ | $\delta_1$ | $\rho_3$ | $\mu_1$ | $\rho_2$ | $\delta_2$ | $\rho_1$ | $\mu_2$ | $\rho_0$ |

The block $(\rho_0 H)(\rho_1 H)$ contains elements from different left cosets: $\rho_0\rho_1 = \rho_1 \in \rho_1 H$, but $\mu_2\rho_1 = \delta_1 \in \rho_3 H$. Both $\rho_0$ and $\mu_2$ represent the same coset $H$, so this demonstrates that changing a representative changes the product coset. Thus coset multiplication is not well defined. There is no coset group here, so it is isomorphic to neither $\mathbb{Z}_4$ nor $V$.

### Exercise 10.9: Left cosets of the half-turn subgroup

Set $K = \{\rho_0, \rho_2\}$. The distinct left cosets are

$$\begin{aligned} K = \{\rho_0, \rho_2\}, \quad \rho_1 K = \{\rho_1, \rho_3\}, \\ \mu_1 K = \{\mu_1, \mu_2\}, \quad \delta_1 K = \{\delta_1, \delta_2\}. \end{aligned}$$

Each pair comes from multiplying the representative by $\rho_0$ and by $\rho_2$, and the four pairs partition the eight elements.

### Exercise 10.10: Right cosets of the half-turn subgroup

The half-turn $\rho_2$ commutes with every square symmetry: algebraically, $r^2 r^i = r^i r^2$ and $r^2(r^i s) = (r^i s)r^2$, using $sr^2 = r^{-2}s = r^2 s$. Hence $aK = Ka$ for each $a \in D_4$. The right cosets are therefore the same four pairs listed in Exercise 9; $K$ is normal.

### Exercise 10.11: Reordered table and the Klein quotient

Using the order $\rho_0, \rho_2, \rho_1, \rho_3, \mu_1, \mu_2, \delta_1, \delta_2$, the table is

| | $\rho_0$ | $\rho_2$ | $\rho_1$ | $\rho_3$ | $\mu_1$ | $\mu_2$ | $\delta_1$ | $\delta_2$ |
|---|---|---|---|---|---|---|---|---|
| $\rho_0$ | $\rho_0$ | $\rho_2$ | $\rho_1$ | $\rho_3$ | $\mu_1$ | $\mu_2$ | $\delta_1$ | $\delta_2$ |
| $\rho_2$ | $\rho_2$ | $\rho_0$ | $\rho_3$ | $\rho_1$ | $\mu_2$ | $\mu_1$ | $\delta_2$ | $\delta_1$ |
| $\rho_1$ | $\rho_1$ | $\rho_3$ | $\rho_2$ | $\rho_0$ | $\delta_1$ | $\delta_2$ | $\mu_2$ | $\mu_1$ |
| $\rho_3$ | $\rho_3$ | $\rho_1$ | $\rho_0$ | $\rho_2$ | $\delta_2$ | $\delta_1$ | $\mu_1$ | $\mu_2$ |
| $\mu_1$ | $\mu_1$ | $\mu_2$ | $\delta_2$ | $\delta_1$ | $\rho_0$ | $\rho_2$ | $\rho_3$ | $\rho_1$ |
| $\mu_2$ | $\mu_2$ | $\mu_1$ | $\delta_1$ | $\delta_2$ | $\rho_2$ | $\rho_0$ | $\rho_1$ | $\rho_3$ |
| $\delta_1$ | $\delta_1$ | $\delta_2$ | $\mu_1$ | $\mu_2$ | $\rho_1$ | $\rho_3$ | $\rho_0$ | $\rho_2$ |
| $\delta_2$ | $\delta_2$ | $\delta_1$ | $\mu_2$ | $\mu_1$ | $\rho_3$ | $\rho_1$ | $\rho_2$ | $\rho_0$ |

Every $2 \times 2$ block lies entirely in a single coset, so multiplication descends to the four cosets. Label them $K, A = \rho_1 K, B = \mu_1 K, C = \delta_1 K$. The induced table is

| | $K$ | $A$ | $B$ | $C$ |
|---|---|---|---|---|
| $K$ | $K$ | $A$ | $B$ | $C$ |
| $A$ | $A$ | $K$ | $C$ | $B$ |
| $B$ | $B$ | $C$ | $K$ | $A$ |
| $C$ | $C$ | $B$ | $A$ | $K$ |

Every nonidentity element has order 2. Thus the quotient is the Klein four-group $V$, not $\mathbb{Z}_4$.

### Exercise 10.12: Index of $\langle 3 \rangle$ in $\mathbb{Z}_{24}$

The multiples of 3 modulo 24 are $0, 3, 6, 9, 12, 15, 18, 21$, so the subgroup has order 8. Lagrange's counting formula gives $(\mathbb{Z}_{24} : \langle 3 \rangle) = 24/8 = 3$. Equivalently the cosets are the three residue classes modulo 3.

### Exercise 10.13: Index of a reflection subgroup in $S_3$

A reflection $\mu_1$ has order 2, so $\langle \mu_1 \rangle = \{e, \mu_1\}$. Since $|S_3| = 6$, its index is $6/2 = 3$.

### Exercise 10.14: Index of a reflection subgroup in $D_4$

The element $\mu_2$ is a reflection and has order 2. Its subgroup has two elements, whereas $D_4$ has eight. The number of cosets is therefore $8/2 = 4$.

### Exercise 10.15: Index of the subgroup generated by $\sigma$

The factors in $\sigma = (1, 2, 5, 4)(2, 3)$ overlap, so their orders cannot be combined by a least-common-multiple rule. Follow their images from right to left:

$$1 \mapsto 2 \mapsto 3 \mapsto 5 \mapsto 4 \mapsto 1.$$

Thus $\sigma = (1, 2, 3, 5, 4)$ has order 5. Therefore $|\langle \sigma \rangle| = 5$ and $(S_5 : \langle \sigma \rangle) = 5!/5 = 24$.

### Exercise 10.16: Index of the subgroup generated by $\mu$

The cycles $(1, 2, 4, 5)$ and $(3, 6)$ are disjoint and have lengths $4, 2$. Hence $\mu$ has order $\operatorname{lcm}(4, 2) = 4$. Its cyclic subgroup has four elements, and the index in $S_6$ is $6!/4 = 720/4 = 180$.

### Exercise 10.17: Definition of a left coset

Require that $H$ be a *subgroup* of $G$, not merely a subset, and that $a \in G$. Then the left coset of $H$ containing $a$ is $aH = \{ah : h \in H\}$. Because $e \in H$, it indeed contains $a = ae$. For an arbitrary subset, the proposed phrase "containing $a$" need not be true.

### Exercise 10.18: Definition of index

The definition using the number of right cosets is correct. The number of left cosets is equal to it, so either convention defines the same index $(G : H)$. For finite $G$ this number is $|G|/|H|$; the definition itself does not require finiteness.

### Exercise 10.19: True or false

(a) True: $aH$ is defined for every subgroup $H \leq G$ and every $a \in G$.

(b) True: $|G| = (G : H)|H|$, so the number of cosets divides $|G|$.

(c) True: a group of prime order is cyclic, and cyclic groups are abelian.

(d) False: the finite subgroup $\{0\}$ of $\mathbb{Z}$ has singleton cosets $n + \{0\}$.

(e) True: $H = eH$.

(f) False: for example $\mathbb{Z} \leq \mathbb{R}$ has cosets $x + \mathbb{Z}$.

(g) True: for $n > 1$, half the permutations are even, so $[S_n : A_n] = 2$.

(h) This is a statement of taste rather than a mathematical proposition; one may agree that the theorem is nice, but there is no objectively defined truth value to prove.

(i) False: $V$ has order 4 but no element of order 4.

(j) True: if $G = \langle g \rangle$ has order $m$ and $d \mid m$, then $g^{m/d}$ has order $d$.

### Exercise 10.20: Different left and right partitions in an abelian group

Impossible. For every $a \in G$ and $h \in H$, commutativity gives $ah = ha$, so $aH = Ha$. Thus the left and right coset partitions must agree for every subgroup of an abelian group.

### Exercise 10.21: A single coset

Take $H = G$, for example $H = G = \mathbb{Z}_6$. Every $aH = aG = G$, so there is exactly one cell in the partition. Conversely an index-1 subgroup must be all of $G$, since its identity coset $H$ is the only cell.

### Exercise 10.22: Six cosets in a group of order six

Take $G = \mathbb{Z}_6$ and the trivial subgroup $H = \{0\}$. Its six cosets are the six singletons, so the partition has six cells. The formula $[G : H] = 6/1$ confirms the count.

### Exercise 10.23: Twelve cosets in a group of order six

Impossible. Every coset is nonempty, so a partition of a six-element set has at most six cells. Equivalently, the index would force $|H| = 6/12 = 1/2$, which is not a possible subgroup order.

### Exercise 10.24: Four cosets in a group of order six

Impossible. Lagrange's formula would give $6 = 4|H|$, so $|H| = 3/2$, not an integer. Thus the index of a subgroup of an order-6 group cannot be 4.

### Exercise 10.25: Synopsis of Lagrange's theorem

The left cosets of $H$ partition finite $G$, and each has $|H|$ elements by the bijection $h \mapsto ah$, so $|G| = [G : H]|H|$ and $|H|$ divides $|G|$.

### Exercise 10.26: The right-coset equivalence relation

The relation is $a \sim_R b$ if and only if $ab^{-1} \in H$. It is reflexive because $aa^{-1} = e \in H$. If $ab^{-1} \in H$, inverse closure gives $(ab^{-1})^{-1} = ba^{-1} \in H$, so it is symmetric. If $ab^{-1} \in H$ and $bc^{-1} \in H$, their product is $ac^{-1} \in H$, proving transitivity. Hence it is an equivalence relation; its class containing $a$ is $Ha$.

### Exercise 10.27: A bijection with a right coset

Define $f : H \to Hg$ by $f(h) = hg$. Its values lie in $Hg$ by definition. If $f(h_1) = f(h_2)$, then $h_1 g = h_2 g$; multiplying on the right by $g^{-1}$ gives $h_1 = h_2$, proving injectivity. Every $x \in Hg$ has the form $hg$ for some $h \in H$, so $x = f(h)$ and the map is surjective. Its inverse is $x \mapsto xg^{-1}$.

### Exercise 10.28: Conjugation condition implies equal cosets

Suppose $g^{-1}hg \in H$ for all $g \in G, h \in H$. If $hg \in Hg$, write $hg = g(g^{-1}hg) \in gH$, so $Hg \subseteq gH$. For the other inclusion replace $g$ by $g^{-1}$ in the hypothesis: $ghg^{-1} \in H$. Then $gh = (ghg^{-1})g \in Hg$, so $gH \subseteq Hg$. The two inclusions give equality.

### Exercise 10.29: Equal coset partitions imply the conjugation condition

Assume the partitions agree. The left coset $gH$ and the right coset $Hg$ both contain $g$, so they must be the same cell of that common partition. For $h \in H$, the element $hg$ is in $Hg = gH$, so $hg = gh'$ for some $h' \in H$. Multiplying on the left by $g^{-1}$ gives $g^{-1}hg = h' \in H$, as required.

### Exercise 10.30: Do equal left cosets imply equal right cosets?

No. In $S_3$, take $H = \{e, (1,2)\}$, $a = (1,3)$, and $b = a(1,2) = (1,2,3)$. Since $b = ah$ for $h \in H$, $aH = bH$. If $Ha = Hb$, the right-coset criterion would give $ba^{-1} \in H$. But

$$ba^{-1} = a(1,2)a^{-1} = (3,2) \notin H.$$

Thus $Ha \neq Hb$, giving a counterexample satisfying the assumed left-coset equality.

### Exercise 10.31: Membership from equal right cosets

True. Since $e \in H$, the element $b = eb$ belongs to $Hb$. The assumption $Ha = Hb$ then gives $b \in Ha$. This conclusion is a membership statement and does not require commutativity or normality.

### Exercise 10.32: Inverting equal cosets

True. If $aH = bH$, take the inverses of every element of both sets. Since inversion is a bijection and $H^{-1} = H$,

$$(aH)^{-1} = H^{-1}a^{-1} = Ha^{-1}, \qquad (bH)^{-1} = Hb^{-1}.$$

Equal sets have equal inverse sets, so $Ha^{-1} = Hb^{-1}$.

### Exercise 10.33: Squaring representatives of equal cosets

False. Use $H = \{e, (1,2)\}$, $a = (1,3)$, and $b = (1,2,3) = a(1,2)$ as in Exercise 30. Then $aH = bH$, but $a^2 = e$, so $a^2H = H$. Meanwhile $b^2 = (1,3,2) \notin H$, so $b^2H \neq H$. Thus squaring a representative need not preserve left-coset equality when the subgroup is not normal.

### Exercise 10.34: Proper subgroups of a group of order $pq$

By Lagrange, the order of a subgroup divides $pq$. If the subgroup is proper, its order is $1, p$, or $q$ (when $p = q$, the list reduces to $1, p$). The trivial group is cyclic. Any group of prime order is cyclic: a nonidentity element generates a subgroup whose order is a divisor greater than 1 of that prime, and so generates the whole group. Hence every proper subgroup is cyclic.

### Exercise 10.35: A bijection between left and right cosets

Define $F$ on left cosets by $F(aH) = Ha^{-1}$. It is well defined because $aH = bH$ implies $Ha^{-1} = Hb^{-1}$, as proved in Exercise 32. Conversely $Ha^{-1} = Hb^{-1}$ implies $aH = bH$ after taking inverse sets, so $F$ is injective. Every right coset $Hg$ equals $F(g^{-1}H)$, proving surjectivity. Thus the collections have the same cardinality even for infinite groups.

### Exercise 10.36: A unique involution in an abelian group of order $2n$

Existence follows by pairing nonidentity elements with their inverses in a group of even order: at least one is its own inverse. For uniqueness, suppose distinct involutions $a, b$ exist. Since the group

is abelian, the four elements $e, a, b, ab$ form a subgroup: all square to $e$, and they are distinct, since $ab = e, a, b$ would force $a = b$, $b = e$, or $a = e$. Thus 4 divides $2n$ by Lagrange, contradicting oddness of $n$. Hence there is exactly one element of order 2.

### Exercise 10.37: No proper nontrivial subgroups

Choose $a \neq e$. Its subgroup $\langle a \rangle$ is nontrivial, so by hypothesis it equals $G$ and $G$ is cyclic. It cannot be infinite cyclic, because then $\langle a^2 \rangle$ is a proper nontrivial subgroup: it contains $a^2 \neq e$ but not $a$. Thus $G$ has a finite order $m > 1$. If $m$ were composite, choose a proper divisor $d$ with $1 < d < m$; the element $a^{m/d}$ generates a subgroup of order $d$, again a contradiction. Therefore $m$ is prime.

### Exercise 10.38: Multiplication of subgroup indices

Let the distinct left cosets of $H$ in $G$ be $a_i H$ for $1 \leq i \leq r$, and those of $K$ in $H$ be $b_j K$ for $1 \leq j \leq s$.

**1. Cover $G$.**

Given $g \in G$, write $g = a_i h$ for some $i$ and $h \in H$. Then $h = b_j k$ for some $j, k \in K$, so $g \in a_i b_j K$.

**2. Show the $rs$ cosets are distinct.**

If $a_i b_j K = a_{i'} b_{j'} K$, representatives differ by a factor in $K \subseteq H$, so $a_i H = a_{i'} H$ and $i = i'$. Cancelling $a_i$ then gives $b_j K = b_{j'} K$, hence $j = j'$. Therefore these are exactly $rs$ distinct left cosets of $K$ in $G$, proving $[G : K] = [G : H][H : K]$. The argument does not require $G$ itself to be finite.

### Exercise 10.39: A subgroup of index two is normal

If $g \in H$, both $gH$ and $Hg$ equal $H$. If $g \notin H$, the two left cosets are $H$ and $gH$, so $gH = G \setminus H$. The right cosets likewise give $Hg = G \setminus H$. Hence $gH = Hg$ for every $g$, proving normality. In fact the same complement argument works without the given finiteness assumption.

### Exercise 10.40: Every element satisfies $a^n = e$

Let the order of $a$ be $d$. By Lagrange applied to $\langle a \rangle$, $d$ divides $n = |G|$, so write $n = dk$. Then $a^n = (a^d)^k = e^k = e$. The order $d$ is finite because $G$ is finite.

### Exercise 10.41: A unique representative in $[0, 1)$

Let a coset be $a + \mathbb{Z}$. Set $x = a - \lfloor a \rfloor$. Since $\lfloor a \rfloor \leq a < \lfloor a \rfloor + 1$, we have $0 \leq x < 1$; subtracting the integer $\lfloor a \rfloor$ also shows $x \in a + \mathbb{Z}$. For uniqueness, if $x, y$ both lie in this coset and in $[0, 1)$, their difference is an integer with $-1 < x - y < 1$. The only such integer is 0, so $x = y$.

### Exercise 10.42: Sine is constant on a coset

The subgroup generated by $2\pi$ in additive $\mathbb{R}$ is $2\pi\mathbb{Z}$. If $y$ belongs to the coset $x + 2\pi\mathbb{Z}$, then $y = x + 2\pi k$ for some integer $k$. By the $2\pi$-periodicity of sine, $\sin y = \sin x$. Thus assigning $x + 2\pi\mathbb{Z} \mapsto \sin x$ is independent of the representative chosen and defines a function on the coset set.

### Exercise 10.43: Double cosets

**(a)** Reflexivity: $a = eae$ with $e \in H, K$. Symmetry: if $a = hbk$, then $b = h^{-1}ak^{-1}$, with the inverse factors still in their subgroups. Transitivity: if $a = h_1bk_1$ and $b = h_2ck_2$, then $a = (h_1h_2)c(k_2k_1)$, with $h_1h_2 \in H$ and $k_2k_1 \in K$. Thus the relation is an equivalence relation. **(b)** Its class containing $a$ consists exactly of all $hak$, with $h \in H$ and $k \in K$, namely the double coset $HaK$. The order of the two factors on the right in the transitivity calculation matters because the group need not be abelian.

### Exercise 10.44: A point stabilizer and its transport coset

**(a)** Let $H = S_{c,c}$. The identity fixes $c$; products of maps fixing $c$ fix $c$; and inverses fix $c$ too. Hence $H \leq S_A$. **(b)** If $d \neq c$, the identity does not send $c$ to $d$, so it is absent from $S_{c,d}$. Therefore this set is not a subgroup. **(c)** Choose a permutation $\tau$ sending $c$ to $d$, for example the transposition $(c, d)$. Then $S_{c,d} = \tau H$. Indeed $(\tau h)(c) = \tau(c) = d$ for $h \in H$. Conversely, if $\sigma(c) = d$, then $(\tau^{-1}\sigma)(c) = c$, so $\tau^{-1}\sigma \in H$ and $\sigma \in \tau H$. Thus it is a left coset of the stabilizer.

### Exercise 10.45: The subgroups of a finite cyclic group

Let $G = \langle g \rangle$ have order $n$. For a divisor $d \mid n$, the element $g^{n/d}$ has order $d$, so $H_d = \langle g^{n/d} \rangle$ is a subgroup of order $d$. Conversely, every subgroup of a cyclic group is cyclic. If $H = \langle g^k \rangle$ has order $d$, then $d = n/\gcd(n, k)$, so $\gcd(n, k) = n/d$. The subgroup generated by $g^k$ is exactly the subgroup generated by $g^{\gcd(n,k)}$: the exponent gcd is an integer combination of $n, k$, and $g^n = e$. Thus $H = \langle g^{n/d} \rangle = H_d$. This proves existence and uniqueness for each divisor, and Lagrange excludes other subgroup orders.

### Exercise 10.46: The divisor sum of the Euler function

Partition the $n$ elements of $G = \langle g \rangle$ according to their exact orders. For each divisor $d \mid n$, there is one subgroup of order $d$, and the elements of exact order $d$ are precisely its generators. A cyclic order-$d$ group has $\varphi(d)$ generators, because its generator powers $h^k$ generate exactly when $\gcd(k, d) = 1$. Therefore counting all elements gives

$$\boxed{\sum_{d|n} \varphi(d) = n.}$$

The divisor $d = 1$ contributes the identity, since $\varphi(1) = 1$.

### Exercise 10.47: A bound on roots forces cyclicity

Put $n = |G|$, and for each divisor $d$ of $n$ let $c_d$ count elements of exact order $d$. If $c_d > 0$, choose such an element $a$. Its cyclic subgroup has $d$ elements, all satisfying $x^d = e$. By the hypothesis there are at most $d$ solutions in all of $G$, so these are all the solutions. Thus every element of exact order $d$ belongs to $\langle a \rangle$, and $c_d = \varphi(d)$. If no such element exists, $c_d = 0$. In either case $c_d \leq \varphi(d)$. Lagrange says every element order divides $n$, so

$$n = \sum_{d|n} c_d \leq \sum_{d|n} \varphi(d) = n.$$

Equality of the two sums of nonnegative terms forces equality in every term: otherwise at least one strictly smaller term and no larger terms would make the sum smaller. In particular $c_n = \varphi(n) > 0$. An element of order $n$ generates all $n$ elements of $G$, so $G$ is cyclic.

# 11 Direct Products and Finitely Generated Abelian Groups

**Conventions.** Permutations compose from right to left. The identity is $e$. A group generated by an element $a$ is denoted $\langle a \rangle$.

### Exercise 11.1: Elements and orders in $\mathbb{Z}_2 \times \mathbb{Z}_4$

The elements are all pairs $(a, b)$ with $0 \le a < 2$ and $0 \le b < 4$. Addition is componentwise. A multiple $k(a, b)$ is zero exactly when both $ka \equiv 0 \pmod 2$ and $kb \equiv 0 \pmod 4$. Hence its order is the least common multiple of the two component orders. The table lists the order of each pair, with first coordinate indexing rows and second coordinate indexing columns:

| $a\backslash b$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| 0 | 1 | 4 | 2 | 4 |
| 1 | 2 | 4 | 2 | 4 |

No element has order 8, so the order-8 group is not cyclic.

### Exercise 11.2: Elements and orders in $\mathbb{Z}_3 \times \mathbb{Z}_4$

The elements are all pairs $(a, b)$ with $0 \le a < 3$ and $0 \le b < 4$. Addition is componentwise. A multiple $k(a, b)$ is zero exactly when both $ka \equiv 0 \pmod 3$ and $kb \equiv 0 \pmod 4$. Hence its order is the least common multiple of the two component orders. The table lists the order of each pair, with first coordinate indexing rows and second coordinate indexing columns:

| $a\backslash b$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| 0 | 1 | 4 | 2 | 4 |
| 1 | 3 | 12 | 6 | 12 |
| 2 | 3 | 12 | 6 | 12 |

The element $(1, 1)$ has order $\operatorname{lcm}(3, 4) = 12$, equal to the group order, so it generates the whole group; the group is cyclic.

### Exercise 11.3: Order of $(2, 6)$

In $\mathbb{Z}_m$, the order of $a$ is $m/\gcd(m, a)$. In the given group $\mathbb{Z}_4 \times \mathbb{Z}_{12}$, the component orders are

$$\frac{4}{\gcd(4, 2)} = 2, \quad \frac{12}{\gcd(12, 6)} = 2.$$

All components return to zero simultaneously at the least common multiple of these orders, so

$$\operatorname{ord}(2, 6) = \operatorname{lcm}(2, 2) = 2.$$

### Exercise 11.4: Order of $(2, 3)$

In $\mathbb{Z}_m$, the order of $a$ is $m/\gcd(m, a)$. In the given group $\mathbb{Z}_6 \times \mathbb{Z}_{15}$, the component orders are

$$\frac{6}{\gcd(6, 2)} = 3, \quad \frac{15}{\gcd(15, 3)} = 5.$$

All components return to zero simultaneously at the least common multiple of these orders, so

$$\operatorname{ord}(2,3) = \operatorname{lcm}(3,5) = 15.$$

### Exercise 11.5: Order of $(8, 10)$

In $\mathbb{Z}_m$, the order of $a$ is $m/\gcd(m,a)$. In the given group $\mathbb{Z}_{12} \times \mathbb{Z}_{18}$, the component orders are

$$\frac{12}{\gcd(12,8)} = 3, \quad \frac{18}{\gcd(18,10)} = 9.$$

All components return to zero simultaneously at the least common multiple of these orders, so

$$\operatorname{ord}(8,10) = \operatorname{lcm}(3,9) = 9.$$

### Exercise 11.6: Order of $(3, 10, 9)$

In $\mathbb{Z}_m$, the order of $a$ is $m/\gcd(m,a)$. In the given group $\mathbb{Z}_4 \times \mathbb{Z}_{12} \times \mathbb{Z}_{15}$, the component orders are

$$\frac{4}{\gcd(4,3)} = 4, \quad \frac{12}{\gcd(12,10)} = 6, \quad \frac{15}{\gcd(15,9)} = 5.$$

All components return to zero simultaneously at the least common multiple of these orders, so

$$\operatorname{ord}(3,10,9) = \operatorname{lcm}(4,6,5) = 60.$$

### Exercise 11.7: Order of $(3, 6, 12, 16)$

In $\mathbb{Z}_m$, the order of $a$ is $m/\gcd(m,a)$. In the given group $\mathbb{Z}_4 \times \mathbb{Z}_{12} \times \mathbb{Z}_{20} \times \mathbb{Z}_{24}$, the component orders are

$$\frac{4}{\gcd(4,3)} = 4, \quad \frac{12}{\gcd(12,6)} = 2, \quad \frac{20}{\gcd(20,12)} = 5, \quad \frac{24}{\gcd(24,16)} = 3.$$

All components return to zero simultaneously at the least common multiple of these orders, so

$$\operatorname{ord}(3,6,12,16) = \operatorname{lcm}(4,2,5,3) = 60.$$

### Exercise 11.8: Largest orders of cyclic subgroups

In $\mathbb{Z}_m \times \mathbb{Z}_n$, every element order divides $\operatorname{lcm}(m,n)$, since that many additions kill both coordinates. The element $(1,1)$ attains this order, so it is the largest cyclic-subgroup order. Thus the answers are

$$\operatorname{lcm}(6,8) = 24, \qquad \operatorname{lcm}(12,15) = 60.$$

These need not equal the group orders 48 and 180, respectively.

### Exercise 11.9: Proper nontrivial subgroups of $\mathbb{Z}_2^2$

All three nonzero elements have order 2. They generate the three subgroups

$$\{(0,0),(1,0)\}, \quad \{(0,0),(0,1)\}, \quad \{(0,0),(1,1)\}.$$

Any subgroup containing two different nonzero vectors contains their sum, the third nonzero vector, and so is the whole group. Thus the three listed subgroups are exactly the proper nontrivial ones.

### Exercise 11.10: Proper nontrivial subgroups of $\mathbb{Z}_2^3$

Regard the group as the three-dimensional vector space over $\mathbb{Z}_2$; subgroup closure is exactly linear-subspace closure, because the only scalars are $0, 1$. The seven subgroups of order 2 are $\{0, v\}$ for

$$v = (1,0,0), (0,1,0), (0,0,1), (1,1,0), (1,0,1), (0,1,1), (1,1,1).$$

The seven subgroups of order 4 are the planes described, respectively, by

$$x = 0, \quad y = 0, \quad z = 0, \quad x + y = 0, \quad x + z = 0, \quad y + z = 0, \quad x + y + z = 0,$$

where arithmetic is modulo 2. Each displayed equation has four solutions. Every two-dimensional subspace is the kernel of one nonzero linear functional, and over $\mathbb{Z}_2$ no two distinct nonzero functionals are scalar multiples. Consequently this lists all seven planes. Proper nontrivial subgroups have dimension 1 or 2, so these fourteen are all of them.

### Exercise 11.11: Order-four subgroups of $\mathbb{Z}_2 imes \mathbb{Z}_4$

An element has order 4 precisely when its second coordinate is odd. There are four such elements, and each cyclic order-4 subgroup has two generators. The two distinct cyclic subgroups are

$$\langle (0,1) \rangle = \{(0,0), (0,1), (0,2), (0,3)\},$$

$$\langle (1,1) \rangle = \{(0,0), (1,1), (0,2), (1,3)\}.$$

Any noncyclic group of order 4 is a Klein group and all its elements are killed by 2. The elements killed by 2 here are exactly

$$\{(0,0), (1,0), (0,2), (1,2)\},$$

which form one more subgroup. This exhausts both possible types of order-4 subgroup.

### Exercise 11.12: Klein subgroups of $\mathbb{Z}_2 imes \mathbb{Z}_4 imes \mathbb{Z}_4$

Every element of a Klein subgroup is killed by 2, so it must lie in

$$T = \{(a, 2b, 2c) : a, b, c \in \mathbb{Z}_2\} \cong \mathbb{Z}_2^3.$$

A Klein subgroup is a two-dimensional subspace of this three-dimensional space. Thus all seven are given by $(a, 2b, 2c)$ subject to one of the conditions

$$a = 0,\ b = 0,\ c = 0,\ a + b = 0,\ a + c = 0,\ b + c = 0,\ a + b + c = 0$$

in $\mathbb{Z}_2$. For example the first is $\{(0,0,0), (0,2,0), (0,0,2), (0,2,2)\}$. Each condition gives four elements, all of exponent 2, hence a Klein group. The seven-plane classification from the preceding exercise proves completeness.

### Exercise 11.13: Direct product descriptions of $\mathbb{Z}_{60}$

A finite direct product of cyclic groups is cyclic precisely when their orders are pairwise relatively prime. Since $60 = 4 \cdot 3 \cdot 5$, the prime-power blocks $4, 3, 5$ must remain intact but can be grouped. Ignoring order of factors, all descriptions using at least two nontrivial factors are

$$\mathbb{Z}_4 \times \mathbb{Z}_{15}, \quad \mathbb{Z}_3 \times \mathbb{Z}_{20}, \quad \mathbb{Z}_5 \times \mathbb{Z}_{12}, \quad \mathbb{Z}_4 \times \mathbb{Z}_3 \times \mathbb{Z}_5.$$

These exhaust the partitions of the three prime-power blocks into at least two groups. If factors $\mathbb{Z}_1$ are permitted, one can insert arbitrarily many trivial factors, giving infinitely many formally different descriptions without changing the group; the finite list uses the usual convention of omitting trivial factors.

### Exercise 11.14: Fill in the numerical blanks

(a) $\operatorname{ord}_{\mathbb{Z}_{24}}(18) = 24/\gcd(24, 18) = 24/6 = 4$.

(b) $|\mathbb{Z}_3 \times \mathbb{Z}_4| = 3 \cdot 4 = 12$.

(c) The orders of $4 \in \mathbb{Z}_{12}$ and $2 \in \mathbb{Z}_8$ are 3 and 4, so $(4, 2)$ has order 12.

(d) $V \cong \mathbb{Z}_2 \times \mathbb{Z}_2$; both blanks are 2.

(e) A finite-order element of $\mathbb{Z}_2 \times \mathbb{Z} \times \mathbb{Z}_4$ must have zero middle coordinate. There are $2 \cdot 1 \cdot 4 = 8$ such elements, all indeed of finite order.

### Exercise 11.15: Maximum order in $\mathbb{Z}_4 times \mathbb{Z}_6$

Every component is killed by its modulus, so $12 = \operatorname{lcm}(4, 6)$ kills every pair. The pair $(1, 1)$ has component orders $4, 6$ and hence order 12. Thus 12 is the maximum possible order.

### Exercise 11.16: Comparing two groups of order 24

Yes. Splitting relatively prime cyclic factors gives

$$\mathbb{Z}_2 \times \mathbb{Z}_{12} \cong \mathbb{Z}_2 \times \mathbb{Z}_4 \times \mathbb{Z}_3, \qquad \mathbb{Z}_4 \times \mathbb{Z}_6 \cong \mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_3.$$

The right-hand products differ only in order of coordinates, so they are isomorphic. An explicit map from the second group to the first is $(a, b) \mapsto (b \bmod 2, c)$, where $c$ is the unique residue modulo 12 satisfying $c \equiv a \pmod 4$, $c \equiv b \pmod 3$; existence and uniqueness follow from coprimality of $4, 3$.

### Exercise 11.17: Maximum order in a three-factor product

For $\mathbb{Z}_8 \times \mathbb{Z}_{10} \times \mathbb{Z}_{24}$ the maximum is $\operatorname{lcm}(8, 10, 24) = 2^3 \cdot 3 \cdot 5 = 120$. Every element is killed by 120, and $(1, 1, 1)$ has exactly that order, proving both the upper bound and its attainment.

### Exercise 11.18: Comparing the groups in Exercise 18

The groups are not isomorphic, although both have the same order and the same maximum element order 120. Count elements killed by 4. In $\mathbb{Z}_m$ the congruence $4x = 0$ has $\gcd(4, m)$ solutions. Therefore the counts are

$$\gcd(4, 8)\gcd(4, 10)\gcd(4, 24) = 4 \cdot 2 \cdot 4 = 32,$$

$$\gcd(4, 4)\gcd(4, 12)\gcd(4, 40) = 4 \cdot 4 \cdot 4 = 64.$$

An isomorphism preserves the equation $4x = 0$ and hence its number of solutions, so the different counts rule out an isomorphism.

### Exercise 11.19: Maximum order in $\mathbb{Z}_4 imes \mathbb{Z}_{18} imes \mathbb{Z}_{15}$

The least common multiple of the factor orders is $\operatorname{lcm}(4, 18, 15) = 2^2 \cdot 3^2 \cdot 5 = 180$. It kills every element, and $(1, 1, 1)$ has that order. Therefore the maximum is 180.

### Exercise 11.20: Comparing the groups in Exercise 20

Yes. Decompose the factors into prime powers:

$$\mathbb{Z}_4 \times \mathbb{Z}_{18} \times \mathbb{Z}_{15} \cong \mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_9 \times \mathbb{Z}_3 \times \mathbb{Z}_5,$$
$$\mathbb{Z}_3 \times \mathbb{Z}_{36} \times \mathbb{Z}_{10} \cong \mathbb{Z}_3 \times \mathbb{Z}_4 \times \mathbb{Z}_9 \times \mathbb{Z}_2 \times \mathbb{Z}_5.$$

The same five prime-power cyclic factors occur, so reordering coordinates gives an isomorphism. Splitting was legitimate because each original factor was split into relatively prime factors.

### Exercise 11.21: Abelian groups of order 8

The classification theorem assigns one group to each partition of 3, using one cyclic factor $\mathbb{Z}_{2^k}$ for each part $k$. Thus the complete list is

$$\mathbb{Z}_8,$$
$$\mathbb{Z}_4 \times \mathbb{Z}_2,$$
$$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2.$$

The partitions are distinct, so the uniqueness in the classification theorem shows the listed groups are pairwise nonisomorphic. There are 3 groups in all.

### Exercise 11.22: Abelian groups of order 16

The classification theorem assigns one group to each partition of 4, using one cyclic factor $\mathbb{Z}_{2^k}$ for each part $k$. Thus the complete list is

$$\mathbb{Z}_{16},$$
$$\mathbb{Z}_8 \times \mathbb{Z}_2,$$
$$\mathbb{Z}_4 \times \mathbb{Z}_4,$$
$$\mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_2,$$
$$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2.$$

The partitions are distinct, so the uniqueness in the classification theorem shows the listed groups are pairwise nonisomorphic. There are 5 groups in all.

### Exercise 11.23: Abelian groups of order 32

The classification theorem assigns one group to each partition of 5, using one cyclic factor $\mathbb{Z}_{2^k}$ for each part $k$. Thus the complete list is

$$\begin{gathered}\mathbb{Z}_{32},\\ \mathbb{Z}_{16}\times\mathbb{Z}_2,\\ \mathbb{Z}_8\times\mathbb{Z}_4,\\ \mathbb{Z}_8\times\mathbb{Z}_2\times\mathbb{Z}_2,\\ \mathbb{Z}_4\times\mathbb{Z}_4\times\mathbb{Z}_2,\\ \mathbb{Z}_4\times\mathbb{Z}_2\times\mathbb{Z}_2\times\mathbb{Z}_2,\\ \mathbb{Z}_2\times\mathbb{Z}_2\times\mathbb{Z}_2\times\mathbb{Z}_2\times\mathbb{Z}_2.\end{gathered}$$

The partitions are distinct, so the uniqueness in the classification theorem shows the listed groups are pairwise nonisomorphic. There are 7 groups in all.

### Exercise 11.24: Abelian groups of order 720

Factor $720 = 2^4 \cdot 3^2 \cdot 5$. The possible 2-primary factors are

$$T \in \{\mathbb{Z}_{16},\ \mathbb{Z}_8\times\mathbb{Z}_2,\ \mathbb{Z}_4\times\mathbb{Z}_4,\ \mathbb{Z}_4\times\mathbb{Z}_2^2,\ \mathbb{Z}_2^4\},$$

and the possible 3-primary factors are $U \in \{\mathbb{Z}_9, \mathbb{Z}_3^2\}$. The 5-primary factor must be $\mathbb{Z}_5$. All groups, with none omitted or repeated up to isomorphism, are therefore

$$\boxed{T\times U\times\mathbb{Z}_5 \quad \text{for every one of the five } T \text{ and two } U.}$$

This specifies $5\cdot 2 = 10$ groups explicitly by independent choices; uniqueness of the prime-power cyclic decomposition distinguishes them.

### Exercise 11.25: Abelian groups of order 1089

Since $1089 = 33^2 = 3^2\cdot 11^2$, each primary component has two possibilities. The four groups are

$$\begin{aligned}&\mathbb{Z}_9\times\mathbb{Z}_{121}, \qquad &&\mathbb{Z}_9\times\mathbb{Z}_{11}^2,\\ &\mathbb{Z}_3^2\times\mathbb{Z}_{121}, \qquad &&\mathbb{Z}_3^2\times\mathbb{Z}_{11}^2.\end{aligned}$$

The exponent-2 partition choices are 2 and $1+1$ for each prime. They produce all primary decompositions and hence all isomorphism types.

### Exercise 11.26: Counts for orders 24, 25, and 600

The printed numbers are 24 and 25, not $2^4$ and $2^5$. Since $24 = 2^3\cdot 3$, the three partitions of 3 give three groups:

$$\mathbb{Z}_8\times\mathbb{Z}_3, \quad \mathbb{Z}_4\times\mathbb{Z}_2\times\mathbb{Z}_3, \quad \mathbb{Z}_2^3\times\mathbb{Z}_3.$$

For order $25 = 5^2$ the two possibilities are $\mathbb{Z}_{25}$ and $\mathbb{Z}_5^2$. Because 24 and 25 are coprime, their primary decompositions can be chosen independently. Hence for $(24)(25) = 600$ there are $3 \cdot 2 = 6$ abelian groups up to isomorphism.

### Exercise 11.27: Multiplicativity for coprime group orders

Suppose $\gcd(m, n) = 1$. The prime factors of $m$ and $n$ are disjoint. By the finite abelian classification theorem, any abelian group of order $mn$ is uniquely determined by its primary factors. Collecting the factors at primes dividing $m$ gives a group of order $m$; collecting those at primes dividing $n$ gives one of order $n$. Conversely, the direct product of any one of the $r$ types of order $m$ and any one of the $s$ types of order $n$ gives a group of order $mn$. Uniqueness of primary decompositions means different pairs give different isomorphism types. Thus there are exactly $rs$.

### Exercise 11.28: Groups of order $10^5$

Factor $10^5 = 2^5 \cdot 5^5$. There are seven partitions of 5, hence seven abelian 2-groups of order $2^5$ and seven abelian 5-groups of order $5^5$. These choices are independent by coprimality, so the answer is $7 \cdot 7 = 49$.

### Exercise 11.29: Partition counts for prime-power orders

**(a)** An abelian group of order $p^n$ corresponds to a partition of $n$: parts $\lambda_1, \ldots, \lambda_k$ give $\mathbb{Z}_{p^{\lambda_1}} \times \cdots \times \mathbb{Z}_{p^{\lambda_k}}$. The required table is

| $n$ | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| number | 2 | 3 | 5 | 7 | 11 | 15 | 22 |

One reproducible way to obtain it is the recurrence $P(n, k) = P(n, k-1) + P(n-k, k)$, splitting partitions according to whether a part $k$ occurs, with $P(0, k) = 1$ and $P(n, 0) = 0$ for $n > 0$; the desired count is $P(n, n)$. **(b)** Distinct primes permit independent choices. Thus

$$\begin{aligned} \text{(i) } p^3q^4r^7 &: 3 \cdot 5 \cdot 15 = 225, \\ \text{(ii) } (qr)^7 = q^7r^7 &: 15 \cdot 15 = 225, \\ \text{(iii) } q^5r^4q^3 = q^8r^4 &: 22 \cdot 5 = 110. \end{aligned}$$

In (iii) the two powers of the same prime $q$ must first be combined; they are not independent primary factors.

### Exercise 11.30: A Cayley digraph on a toroidal grid

Place a vertex $(i, j)$ in an $m$-by-$n$ grid, with $0 \le i < m$, $0 \le j < n$. Draw one solid arrow

$$(i, j) \longrightarrow (i + 1 \bmod m, j)$$

for the generator $(1, 0)$, and one dashed arrow

$$(i, j) \dashrightarrow (i, j + 1 \bmod n)$$

for $(0,1)$. Thus each row wraps into an $m$-cycle and each column into an $n$-cycle. A corner of the schematic is

$$\begin{array}{ccccc} (0,2) & \longrightarrow & (1,2) & \longrightarrow & \cdots \\ \uparrow & & \uparrow & & \\ (0,1) & \longrightarrow & (1,1) & \longrightarrow & \cdots \\ \uparrow & & \uparrow & & \\ (0,0) & \longrightarrow & (1,0) & \longrightarrow & \cdots \end{array}$$

Vertical arrows represent the second arc type; include the wrap arrows from $i = m-1$ to $i = 0$ and from $j = n-1$ to $j = 0$. These rules specify every labeled vertex and arc, including the boundary identifications.

### Exercise 11.31: Two concentric polygon digraphs

Let $r$ denote the solid-arrow generator, of order $n$, and $s$ the dashed-edge generator, of order 2. Label one polygon $e, r, \ldots, r^{n-1}$ and the matching vertices in the other $s, rs, \ldots, r^{n-1}s$. **(a)** The group is abelian exactly when the arrows on the two polygons run in the same direction. In that case traversing a dashed edge before or after a solid edge reaches the same vertex, so $rs = sr$; because $r, s$ generate the group, this proves commutativity. Opposite directions instead give $sr = r^{-1}s \neq rs$ for $n \geq 3$. **(b)** In the abelian case all $2n$ elements are uniquely $r^i s^j$ with $0 \leq i < n$, $j = 0, 1$, and multiplication adds both exponents. Thus $G \cong \mathbb{Z}_n \times \mathbb{Z}_2$. **(c)** This product is cyclic exactly when $\gcd(n, 2) = 1$, that is, $n$ is odd. **(d)** Opposite arrow directions give exactly the usual dihedral relations, and hence the $2n$-element group is $D_n$.

### Exercise 11.32: True or false: direct products

(a) True: $(g_1, g_2) \mapsto (g_2, g_1)$ is an isomorphism.

(b) Componentwise computation is precisely the multiplication rule; the description "easy" is informal, but the intended mathematical content is true.

(c) False: $\mathbb{Z} \times \mathbb{Z}$ is a direct product of infinite groups.

(d) True: the orders of two proper nontrivial finite factors would both exceed 1 and multiply to the prime group order, which is impossible.

(e) False: every element of $\mathbb{Z}_2 \times \mathbb{Z}_4$ has order dividing 4, unlike a generator of $\mathbb{Z}_8$.

(f) False: the product has 8 elements, while $S_8$ has 8!; it is also abelian.

(g) False: both groups have 24 elements, but $\mathbb{Z}_3 \times \mathbb{Z}_8$ is abelian and $S_4$ is not.

(h) False: the identity has order 1; other orders can also be 2 or 4.

(i) False: $|\mathbb{Z}_{12} \times \mathbb{Z}_{15}| = 12 \cdot 15 = 180$.

(j) True: there are independently $m$ choices in the first coordinate and $n$ in the second.

### Exercise 11.33: An indecomposable abelian group

Take $G = \mathbb{Z}_p$ for a prime $p$. Its only subgroups are $\{0\}$ and itself, so there are no proper nontrivial subgroups to use as internal factors. Alternatively, an internal product of two nontrivial finite groups would have composite order, contrary to $|G| = p$.

### Exercise 11.34: Subgroups isomorphic to the full product

**(a)** Exactly one. The finite group $\mathbb{Z}_5 \times \mathbb{Z}_6$ has 30 elements, and a subgroup isomorphic to it must also have 30 elements, hence equal the whole group. **(b)** Infinitely many: for each positive integer $k$, $k\mathbb{Z} \times \mathbb{Z}$ is a subgroup of $\mathbb{Z}^2$ isomorphic to $\mathbb{Z}^2$ via $(a, b) \mapsto (ka, b)$. These subgroups are distinct. More precisely there are countably infinitely many, since any subgroup isomorphic to $\mathbb{Z}^2$ has two generators in the countable set $\mathbb{Z}^2$, and there are only countably many ordered pairs of possible generators.

### Exercise 11.35: A composite-order group with no nontrivial direct decomposition

Take $\mathbb{Z}_4$. Its only proper nontrivial subgroup is $\{0, 2\}$. Any two proper nontrivial internal factors would therefore coincide and have nontrivial intersection, so cannot give an internal direct product. Equivalently the only possible nontrivial factor orders are $2, 2$, whose external product is $\mathbb{Z}_2^2$, with no element of order 4, and hence is not $\mathbb{Z}_4$.

### Exercise 11.36: True or false: classification of abelian groups

(a) True: every group of prime order is cyclic.

(b) False: $\mathbb{Z}_2^2$ has prime-power order 4 and is not cyclic.

(c) False: every sum of 4's and 6's modulo 8 is even, so these generate only $\{0, 2, 4, 6\}$.

(d) True: $5 - 4 = 1$, so $\{4, 5, 6\}$ generates 1 and hence $\mathbb{Z}_8$.

(e) True: a finite group is finitely generated, so the theorem applies.

(f) False: $\mathbb{Z}$ and $\mathbb{Z} \times \mathbb{Z}_2$ both have Betti number 1, but only the latter has nontrivial torsion.

(g) True: the primary decomposition contains a 5-power cyclic factor, which has an element of order 5.

(h) False: $\mathbb{Z}_2^2$ has order divisible by 4 but no element of order 4.

(i) True: the primary decomposition contains elements of orders 2 and 3 in different primary factors; their product has order 6.

(j) True: a nonzero free $\mathbb{Z}$ factor would make the group infinite, so a finite group's Betti number is 0.

### Exercise 11.37: Changing the prime in a prime-power order

The numbers are equal. For either prime, the isomorphism types are parametrized by the same partitions of the exponent $r$. Replacing every factor $\mathbb{Z}_{p^k}$ in a decomposition by $\mathbb{Z}_{q^k}$ gives a bijection

of the two lists of isomorphism types. Thus the count depends on $r$ alone, not on the prime.

### Exercise 11.38: Order-eight and order-four subgroups of an order-72 group

Write $G = P \times Q$ with $|P| = 8$ and $|Q| = 9$, using the primary decomposition. **(a)** There is exactly one subgroup of order 8, namely $P \times \{0\}$. An element of a group of order 8 has order a power of 2, so its projection into the order-9 group $Q$ is zero. Thus any order-8 subgroup lies in $P \times \{0\}$ and must equal it. **(b)** The number of order-4 subgroups is not determined by $|G|$ alone. If $P = \mathbb{Z}_8$, it has exactly one order-4 subgroup. If $P = \mathbb{Z}_2^3$, it has seven planes of order 4. Taking, for example, $Q = \mathbb{Z}_9$ gives two groups of order 72 with different answers.

### Exercise 11.39: The torsion subgroup

Let $T$ consist of elements of finite order. The identity is in $T$. If $a^r = e$ and $b^s = e$, commutativity gives $(ab)^{rs} = a^{rs}b^{rs} = e$, so $ab$ has finite order and belongs to $T$. If $a^r = e$, then $(a^{-1})^r = (a^r)^{-1} = e$, so $a^{-1} \in T$. Thus $T$ is a subgroup. The commutativity assumption is used in expanding the power of $ab$; it cannot simply be omitted.

### Exercise 11.40: Orders of two torsion subgroups

In a product with a $\mathbb{Z}$ coordinate, finite order forces that coordinate to be 0, because a nonzero integer has infinite additive order. All elements in the remaining finite factors are torsion. Hence the torsion subgroups are

$$\mathbb{Z}_4 \times \{0\} \times \mathbb{Z}_3 \quad \text{of order } 12,$$

$$\mathbb{Z}_{12} \times \{0\} \times \mathbb{Z}_{12} \quad \text{of order } 144.$$

### Exercise 11.41: Torsion in the nonzero real numbers

If $x^n = 1$ for a real nonzero $x$, then $|x|^n = 1$, so $|x| = 1$ and $x = \pm 1$. Conversely 1 has order 1 and $-1$ has order 2. Thus the torsion subgroup is $\{1, -1\} \cong \mathbb{Z}_2$.

### Exercise 11.42: Torsion in the nonzero complex numbers

A complex number has finite multiplicative order exactly when it is a root of unity. Indeed $z^n = 1$ implies $z = e^{2\pi ik/n}$ for some integer $k$, and any such number has a finite power equal to 1. Therefore

$$T = \{e^{2\pi iq} : q \in \mathbb{Q}\},$$

the group of all roots of unity. It is isomorphic to $\mathbb{Q}/\mathbb{Z}$ under the map $q + \mathbb{Z} \mapsto e^{2\pi iq}$.

### Exercise 11.43: Torsion and free parts of a finitely generated abelian group

By the classification theorem choose an isomorphism $G \cong F \times \mathbb{Z}^r$, where $F$ is a finite product of finite cyclic prime-power groups. An element has finite order precisely when all its $\mathbb{Z}$ coordinates are zero, so the torsion subgroup corresponds to $F \times \{0\}$. The subgroup corresponding to $\{e\} \times \mathbb{Z}^r$ is torsion

free. These subgroups intersect trivially, commute elementwise, and every element is their product; transporting them back through the isomorphism gives an internal direct-product decomposition of $G$. This includes the case of a trivial finite or free part.

### Exercise 11.44: Torsion coefficients

**(a)** Since $\gcd(4, 9) = 1$, $\mathbb{Z}_4 \times \mathbb{Z}_9 \cong \mathbb{Z}_{36}$. The only torsion coefficient is 36. **(b)** The group is $\mathbb{Z}_6 \times \mathbb{Z}_{12} \times \mathbb{Z}_{20}$. Split into prime-power factors and sort the exponents at each prime, padding with zero exponents on the left:

| $p$ | first | second | third |
|---|---|---|---|
| 2 | 1 | 2 | 2 |
| 3 | 0 | 1 | 1 |
| 5 | 0 | 0 | 1 |

Combining each column gives 2, $2^2 \cdot 3 = 12$, and $2^2 \cdot 3 \cdot 5 = 60$. Hence the torsion coefficients are $2, 12, 60$, and $2 \mid 12 \mid 60$. **(c)** For an arbitrary finite product, factor all moduli into prime powers. For each prime put the exponents in nondecreasing order and pad shorter lists with zero exponents at their left ends to obtain equal lengths. Multiply the prime powers column by column. The resulting column products divide the next ones, because every prime exponent is nondecreasing. Reordering primary factors and combining relatively prime factors proves this process preserves the isomorphism type. Discard any leading coefficient 1.

### Exercise 11.45: Synopsis of the coprime-product criterion

The element $(1, 1) \in \mathbb{Z}_m \times \mathbb{Z}_n$ has order $\operatorname{lcm}(m, n)$, so when $\gcd(m, n) = 1$ it generates the entire order-$mn$ group. If the gcd is $d > 1$, every element is killed by $mn/d < mn$, so no element can generate the whole group and the product is not cyclic.

### Exercise 11.46: A direct product of abelian groups is abelian

Let $a = (a_1, \ldots, a_r)$ and $b = (b_1, \ldots, b_r)$ belong to the product. Since each factor is abelian, $a_i b_i = b_i a_i$ for every $i$. Therefore

$$ab = (a_1 b_1, \ldots, a_r b_r) = (b_1 a_1, \ldots, b_r a_r) = ba.$$

This is exactly commutativity of the product operation.

### Exercise 11.47: Elements of order at most two

The given set is $H = \{g \in G : g^2 = e\}$. It contains $e$. If $a, b \in H$, commutativity gives $(ab)^2 = a^2 b^2 = e$, so $ab \in H$. Also $(a^{-1})^2 = (a^2)^{-1} = e$. Thus $H$ is a subgroup. Any nonidentity element killed by 2 has order exactly 2, so this is precisely the set described in the question.

### Exercise 11.48: Identity together with all elements of an exact order

For order 3, the answer is yes: the set is $\{g : g^3 = e\}$, and commutativity shows it is closed under products and inverses. For order 4, the answer is no. In $\mathbb{Z}_4$, 1 has order 4 but $1 + 1 = 2$ has order 2, so the set $\{0, 1, 3\}$ is not closed. In general the universal property holds exactly for $n = 1$ and for prime

$n$. For prime $p$, every nonidentity solution of $g^p = e$ has order $p$, so the same kernel-style closure argument works. If $n$ is composite, choose a prime $p \mid n$ with $p < n$ and work in $\mathbb{Z}_n \times \mathbb{Z}_n$. Both

$$a = (1, 0), \qquad b = (-1, n/p)$$

have order $n$ because their first coordinates do. But $a + b = (0, n/p)$ has order $p$, which is neither 1 nor $n$. Thus the prescribed set fails closure. This also explains the distinction between exact order $n$ and order dividing $n$.

### Exercise 11.49: Why commutativity is needed for involutions

In the nonabelian group $S_3$, the identity and three transpositions are exactly the elements of order 1 or 2. But $(1, 2)(2, 3) = (1, 2, 3)$ has order 3 and does not lie in that set. Hence it is not a subgroup, providing a counterexample when the abelian hypothesis is removed.

### Exercise 11.50: The coordinate subgroups in an external product

Write the two coordinate subgroups as $\widetilde{H} = H \times \{e_K\}$ and $\widetilde{K} = \{e_H\} \times K$. **(a)** Every $(h, k)$ equals $(h, e_K)(e_H, k)$. **(b)** Componentwise multiplication gives

$$(h, e_K)(e_H, k) = (h, k) = (e_H, k)(h, e_K),$$

so the two subgroups commute elementwise even if the individual groups are nonabelian. **(c)** An element in both coordinate subgroups has first coordinate $e_H$ and second coordinate $e_K$, so the intersection is exactly $\{(e_H, e_K)\}$.

### Exercise 11.51: Recognizing an internal direct product

By the first assumed condition, every $g \in G$ is $hk$. If $hk = h'k'$, rearrange to obtain

$$h'^{-1}h = k'k^{-1} \in H \cap K = \{e\}.$$

Thus $h = h'$ and $k = k'$, proving uniqueness. Define $\Phi : H \times K \to G$ by $\Phi(h, k) = hk$. Existence and uniqueness say it is bijective. Since every element of $H$ commutes with every element of $K$,

$$\Phi(h, k)\Phi(h', k') = hkh'k' = hh'kk' = \Phi(hh', kk').$$

It is therefore a homomorphism and an isomorphism, making precise the claimed structural identity.

### Exercise 11.52: Detecting a noncyclic finite abelian group

Suppose first $G$ contains $\mathbb{Z}_p^2$. This subgroup is not cyclic: every nonidentity element has order $p$, while the group has order $p^2$. Every subgroup of a cyclic group is cyclic, so $G$ cannot be cyclic. Conversely, decompose finite abelian $G$ into its prime-power cyclic factors. If each prime occurred in at most one factor, these factor orders would be pairwise coprime, and their product would be cyclic. Since $G$ is not cyclic, some prime $p$ occurs in at least two factors, say $\mathbb{Z}_{p^a}$ and $\mathbb{Z}_{p^b}$. Their order-$p$ subgroups generated by $p^{a-1}$ and $p^{b-1}$ have direct product $\mathbb{Z}_p^2$, which embeds in $G$. This proves the equivalence.

### Exercise 11.53: Element orders in a prime-power group

If $|G| = p^r$, then for any $g \in G$, Lagrange's theorem says $\operatorname{ord}(g)$ divides $p^r$. The positive divisors are exactly $p^s$ for $0 \leq s \leq r$, so every element order is a power of $p$ (including $p^0 = 1$). This proof did not use commutativity; therefore the abelian hypothesis can be dropped.

### Exercise 11.54: Cancellation for finitely generated abelian groups

By classification, a finitely generated abelian group is specified up to isomorphism by its free rank and, for every prime $p$ and exponent $j \geq 1$, the multiplicity of the factor $\mathbb{Z}_{p^j}$. In a direct product these ranks and multiplicities add. The isomorphism $G \times K \cong H \times K$ therefore gives

$$r_G + r_K = r_H + r_K, \qquad m_G(p, j) + m_K(p, j) = m_H(p, j) + m_K(p, j).$$

All quantities are finite nonnegative integers, since the groups are finitely generated. Cancelling the common terms yields equal free ranks and all equal prime-power multiplicities for $G, H$. The uniqueness part of the theorem then gives $G \cong H$.

# 12 Plane Isometries

**Conventions.** Permutations compose from right to left. The identity is $e$. A group generated by an element $a$ is denoted $\langle a \rangle$.

### Exercise 12.1: Symmetries depend on ambient dimension

(a) For a point $c \in \mathbb{R}$, the isometries fixing it are $x \mapsto x$ and $x \mapsto 2c - x$, the reflection through $c$. An isometry of the line has form $x \mapsto x + t$ or $x \mapsto -x + t$, and the condition of fixing $c$ determines $t$ in each case.

(b) For a point $P$ in the plane, all rotations about $P$ and all reflections across lines through $P$ fix it, together with the identity. There are infinitely many, unlike in the line. Translations and genuine glide reflections fix no points.

(c) A nondegenerate segment in $\mathbb{R}$ has only the identity and reflection through its midpoint: its two endpoints must either both stay fixed or be interchanged.

(d) In $\mathbb{R}^2$ the symmetries are the identity, reflection in the line containing the segment, reflection in its perpendicular bisector, and the half-turn about its midpoint. They form a Klein four-group. Fixing both endpoints allows reflection in the supporting line, and interchanging them allows the other reflection or the half-turn.

(e) In $\mathbb{R}^3$ examples include every rotation about the supporting line, reflection in any plane containing that line, and reflection in the perpendicular midpoint plane. More fully, with midpoint at the origin and segment on the $x$-axis, all maps $(x, y, z) \mapsto (\varepsilon x, Q(y, z))$ with $\varepsilon = \pm 1$ and $Q$ any orthogonal $2 \times 2$ matrix preserve the segment.

### Exercise 12.2: Orientation under composition

Assign sign $+1$ to an orientation-preserving map $P$ and $-1$ to an orientation-reversing map $R$. Under composition these signs multiply, as can be seen from the determinants of the orthogonal linear parts of the isometries. Thus

| | $P$ | $R$ |
|---|---|---|
| $P$ | $P$ | $R$ |
| $R$ | $R$ | $P$ |

In particular two reversals restore orientation, while one reversal changes it.

### Exercise 12.3: Types of products of plane isometries

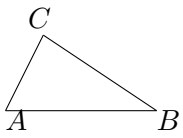
Use $t$ for a nonzero translation, $p$ for a nonidentity rotation, $\mu$ for a reflection, and $\gamma$ for a genuine glide reflection. With identity outcomes omitted, every possible type is recorded by

| | $t$ | $p$ | $\mu$ | $\gamma$ |
|---|---|---|---|---|
| $t$ | $t$ | $p$ | $\mu, \gamma$ | $\mu, \gamma$ |
| $p$ | $p$ | $t, p$ | $\mu, \gamma$ | $\mu, \gamma$ |
| $\mu$ | $\mu, \gamma$ | $\mu, \gamma$ | $t, p$ | $t, p$ |
| $\gamma$ | $\mu, \gamma$ | $\mu, \gamma$ | $t, p$ | $t, p$ |

Here is why the lists are exhaustive. Orientation rules eliminate $t, p$ from a product with exactly one reversing factor, and eliminate $\mu, \gamma$ from a product with zero or two reversing factors. Translations compose by adding vectors. A translation times a rotation keeps the same nonzero rotational angle, hence is another rotation. Two rotations can add to a nonzero angle (rotation) or cancel their angles about different centers (translation). Two reflections across intersecting lines give a rotation; across distinct parallel lines they give a translation. Adding a translation to an orientation-reversing map can introduce or cancel displacement along its axis, giving either a reflection or a glide reflection. Choosing intersecting versus parallel axes for products involving reversing factors gives respectively the rotation and translation alternatives. These constructions realize every entry; inverse choices may give the omitted identity.

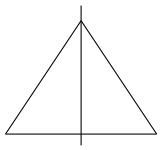
### Exercise 12.4: A figure with only the identity symmetry

Take a scalene triangle, for example vertices $A = (0, 0)$, $B = (4, 0)$, $C = (1, 2)$. Its three side lengths are $4, \sqrt{5}, \sqrt{13}$, all distinct. Any symmetry must preserve side lengths, so it fixes all three vertices and is the identity. A sketch is



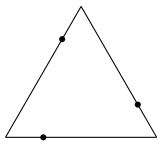

### Exercise 12.5: A figure with two symmetries

Use an isosceles triangle that is not equilateral, such as vertices $(-2, 0), (2, 0), (0, 3)$. The unique vertex between the two equal sides must be fixed; the base endpoints can either stay fixed or interchange. Thus the only symmetries are the identity and reflection across the vertical altitude, forming $\mathbb{Z}_2$.



### Exercise 12.6: A figure with three symmetries

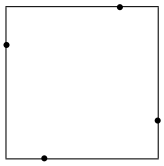
Draw an equilateral triangle and mark a heavy dot one quarter of the way along each edge, always proceeding counterclockwise around the boundary. The dots are part of the figure.



The rotations through $0°, 120°, 240°$ preserve the marks. A reflection reverses the boundary direction and would put the marks three quarters of the way along each counterclockwise edge, where there are no marks. Since the convex hull is the equilateral triangle, every possible symmetry was already among its rotations and reflections. Thus exactly the three rotations remain, giving $\mathbb{Z}_3$.

### Exercise 12.7: A figure with cyclic order-four symmetry

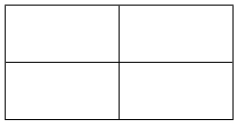
Use a square with one heavy dot one quarter of the way along each edge, in a consistent counterclockwise direction:



All four quarter-turn rotations preserve the figure. Every square reflection reverses the direction along its boundary and moves each quarter-edge marker to a missing three-quarter-edge marker. Thus none is a symmetry. The convex hull forces every symmetry to be a square symmetry, so the group consists of the four rotations and is $\mathbb{Z}_4$.

### Exercise 12.8: A figure with Klein-four symmetry

A rectangle with unequal side lengths has the identity, reflection in each of its two midlines, and the half-turn about its center. A quarter-turn would interchange its unequal dimensions, so is not allowed. Each of its three nonidentity symmetries has order 2, and the two reflections multiply to the half-turn, giving the Klein group $V$.



### Exercise 12.9: Orders of the four isometry types

A nonzero translation has infinite order, because its $k$th power translates by $k$ times its nonzero vector. A rotation through angle $\theta$ has finite order $q \geq 2$ when $\theta/(2\pi) = p/q$ in lowest terms, and infinite order when this ratio is irrational. Every reflection has order 2. A genuine glide reflection has infinite order: if its nonzero displacement along the axis is $v$, its square is the nonzero translation by $2v$. An element whose square has infinite order cannot have finite order.

### Exercise 12.10: Which types have fixed points?

A nonidentity rotation fixes its center. A reflection fixes every point of its axis. A nonzero translation has no fixed point because $P + v = P$ would force $v = 0$. A genuine glide reflection also has none: its component parallel to the axis moves every point by the same nonzero amount, which cannot vanish. Thus exactly rotations and reflections can have fixed points.

### Exercise 12.11: Exactly one fixed point

Only a nonidentity rotation. Its center is fixed, and any other point moves around a circle by a nonzero angle. A reflection has an entire line of fixed points, while translations and genuine glide reflections have none.

### Exercise 12.12: Exactly two fixed points

None of the four nonidentity types. A rotation has one, a reflection has infinitely many, and translations and glide reflections have none. The identity fixes the whole plane and also does not have exactly two.

### Exercise 12.13: Infinitely many fixed points

Among the four nonidentity types, only reflections: their fixed-point set is their entire axis. If the identity is also allowed, it fixes every point of the plane and therefore is another isometry with infinitely many fixed points.

### Exercise 12.14: Three noncollinear fixed points force the identity

Let $A, B, C$ be the fixed noncollinear points and let $X$ be any point. An isometry preserves its distances to all three: the image $Y$ must satisfy $YA = XA$, $YB = XB$, and $YC = XC$. The first two circles meet in at most two points, which, when distinct, are mirror images across $AB$. Since $C$ is off $AB$, those two mirror points have different distances to $C$; hence the third distance selects $X$ uniquely. Thus $Y = X$ for every $X$ and the isometry is the identity. This is the geometric reason three noncollinear distance references determine a plane point.

### Exercise 12.15: Isometries agreeing on three points

Suppose $\phi(P_i) = \psi(P_i)$ at the three given noncollinear points. Form the isometry $\eta = \psi^{-1}\phi$. Then $\eta(P_i) = \psi^{-1}(\psi(P_i)) = P_i$ for each $i$. By Exercise 14, $\eta$ is the identity. Multiplying $\psi^{-1}\phi = \mathrm{id}$ on the left by $\psi$ yields $\phi = \psi$.

### Exercise 12.16: All rotations together with the identity

They do not form a subgroup. Let $r_A(X) = 2A - X$ and $r_B(X) = 2B - X$ be half-turns about two distinct centers. Then

$$(r_B r_A)(X) = 2B - (2A - X) = X + 2(B - A),$$

a nonzero translation. This product is neither a rotation nor the identity, so closure fails.

### Exercise 12.17: Translations together with the identity

Yes. Write $T_v(X) = X + v$, allowing $v = 0$ for the identity. Then $T_v T_w = T_{v+w}$ and $T_v^{-1} = T_{-v}$. The sums and negatives of plane vectors are plane vectors, so the set contains the identity and is closed

under products and inverses.

### Exercise 12.18: Rotations about a common point

Yes. After translating coordinates so that the fixed center is the origin, composition of rotations adds angles, and inversion negates the angle. Thus $R_\alpha R_\beta = R_{\alpha+\beta}$ and $R_\alpha^{-1} = R_{-\alpha}$, with angles read modulo $2\pi$. Angle 0 gives the identity, so these maps form a subgroup.

### Exercise 12.19: One reflection together with the identity

Yes. If $s$ is reflection across the specified line, then $s^2 = e$. Hence $\{e, s\}$ is closed under multiplication, and each element is its own inverse. This is a subgroup isomorphic to $\mathbb{Z}_2$.

### Exercise 12.20: All glide reflections together with the identity

No. A genuine glide reflection $g$ has nonzero displacement $v$ along its axis, and $g^2$ is translation by $2v$. This is neither a genuine glide reflection nor the identity. Thus the product of an allowed element with itself leaves the proposed set.

### Exercise 12.21: Isometries in finite subgroups

Every element of a finite group has finite order. Nonzero translations and genuine glide reflections have infinite order and cannot occur. Reflections can occur with order 2, and rotations can occur when their angle is a rational multiple of $2\pi$. Thus the possible nonidentity types are precisely reflections and finite-order rotations.

### Exercise 12.22: The rotation subgroup of a finite plane group

Restrict the orientation sign map to $G$: its value is $+1$ on preserving elements and $-1$ on reversing elements, and signs multiply. Its $+1$ set $H$ is therefore closed under products and inverses and contains the identity. In a finite group, the preceding exercise excludes nonzero translations, so $H$ consists exactly of the rotations and identity described in the question. If there is no reversing element, $H = G$. Otherwise fix a reversing $s \in G$; multiplication by $s$ is a bijection from $H$ to the reversing elements, because it changes the sign and has inverse multiplication by $s^{-1}$. Hence $|G| = 2|H|$.

### Exercise 12.23: A finite group of rotations about a point is cyclic

If $G = \{e\}$, it is already cyclic. Otherwise choose the smallest positive angle $\theta \in (0, 2\pi)$ occurring among its finitely many rotations, and let $r$ be that rotation. For any rotation angle $\alpha \in [0, 2\pi)$, division by $\theta$ writes $\alpha = q\theta + \delta$ with $0 \le \delta < \theta$. The rotation $r^{-q}R_\alpha$ lies in $G$ and has angle $\delta$. Minimality of $\theta$ forces $\delta = 0$, so every element is a power of $r$. This proves $G = \langle r \rangle$. Since its order is finite, $r$ also has finite order and its multiples close up exactly around the circle.

### Exercise 12.24: Repeated F figures

**Convention.** Here "nontrivial glide" means one not expressible as a product of a translation and reflection that are themselves symmetries, as defined on printed page 116.

(a) No.

(b) No.

(c) No.

(d) No.

(e) The group is isomorphic to $\mathbb{Z}$.

The upper and lower parts of F differ, and its left and right sides differ. Only integral translations by the spacing preserve the repeated upright figures.

### Exercise 12.25: Repeated T figures

**Convention.** Here "nontrivial glide" means one not expressible as a product of a translation and reflection that are themselves symmetries, as defined on printed page 116.

(a) No.

(b) No.

(c) Yes.

(d) No.

(e) The group is isomorphic to $D_\infty$.

Each T has a vertical mirror, but no horizontal mirror or half-turn symmetry. A unit translation $t$ and a vertical reflection $s$ generate the group, with $s^2 = e$ and $sts = t^{-1}$.

### Exercise 12.26: Repeated E figures

**Convention.** Here "nontrivial glide" means one not expressible as a product of a translation and reflection that are themselves symmetries, as defined on printed page 116.

(a) No.

(b) Yes.

(c) No.

(d) No.

(e) The group is isomorphic to $\mathbb{Z} \times \mathbb{Z}_2$.

Horizontal reflection through the middle bar preserves every E; vertical reflection and half-turn do not. This reflection commutes with the unit translation. Glides obtained by combining these two symmetries are trivial in the book's sense.

### Exercise 12.27: Repeated Z figures

**Convention.** Here "nontrivial glide" means one not expressible as a product of a translation and reflection that are themselves symmetries, as defined on printed page 116.

(a) Yes.

(b) No.

(c) No.

(d) No.

(e) The group is isomorphic to $D_\infty$.

A half-turn about the center of a Z preserves it, but neither horizontal nor vertical reflection does. The half-turn $r$ reverses translation: $rtr = t^{-1}$, yielding the infinite dihedral group.

### Exercise 12.28: Repeated H figures

**Convention.** Here "nontrivial glide" means one not expressible as a product of a translation and reflection that are themselves symmetries, as defined on printed page 116.

(a) Yes.

(b) Yes.

(c) Yes.

(d) No.

(e) The group is isomorphic to $D_\infty \times \mathbb{Z}_2$.

The H has both horizontal and vertical mirrors and a half-turn. Translation and a vertical mirror generate $D_\infty$, while the horizontal mirror is an independent commuting involution. Every glide is the product of an existing translation and a horizontal reflection.

### Exercise 12.29: Alternating bent strokes

**Convention.** Here "nontrivial glide" means one not expressible as a product of a translation and reflection that are themselves symmetries, as defined on printed page 116.

(a) No.

(b) No.

(c) No.

(d) Yes.

(e) The group is isomorphic to $\mathbb{Z}$.

The upper and lower bent strokes are horizontal mirror images displaced by half a repeat. A glide $g$ through that half spacing interchanges the two rows, and $g^2$ is the full translation. No pure reflection or half-turn matches the direction of the short horizontal stubs. All symmetries are powers of $g$.

### Exercise 12.30: Alternating cup and cap figures

**Convention.** Here "nontrivial glide" means one not expressible as a product of a translation and reflection that are themselves symmetries, as defined on printed page 116.

(a) Yes.

(b) No.

(c) Yes.

(d) Yes.

(e) The group is isomorphic to $D_\infty$.

Vertical lines through cup or cap centers are mirrors. A half-period glide $g$ in the horizontal baseline interchanges cups and caps; its reflection component alone is not a symmetry. Together with a vertical reflection $s$, it gives $sgs = g^{-1}$, so the group is infinite dihedral. Their product also yields half-turns.

### Exercise 12.31: The square grid

**(a)** Rotations through $90°$ and $180°$ occur, about square centers or appropriate grid vertices; no smaller positive angle is possible because the two perpendicular edge directions must be preserved as a set. **(b)** Yes: horizontal, vertical, and diagonal mirror lines preserve the grid. **(c)** No nontrivial glides in the book's sense. A symmetry has the form $X \mapsto AX + v$ with $v$ a lattice vector and $A \in D_4$ a symmetry fixing a grid vertex. If it is a glide, $A$ is a reflection, so it is already a product of a translation and a reflection in the symmetry group. Ordinary glides do exist; the word "nontrivial" is essential here.

### Exercise 12.32: The staggered square or brick pattern

**(a)** A half-turn is a symmetry, but a quarter-turn is not: vertical grid lines continue indefinitely, whereas horizontal edges terminate at the staggered joins. Thus the only angle in $(0, 180°]$ is $180°$. **(b)** Yes. Take unit square tiles in column $k$ shifted vertically by $k/2$. There are horizontal mirrors at integer or half-integer heights and vertical mirrors through column centers. **(c)** No nontrivial glides under the book's product definition. The group is generated by its centered rectangular translation lattice and the two perpendicular reflections through a column center on a horizontal mirror. Every glide has one of those reflection linear parts and differs from such a reflection by a translation already in the group. The translation/reflection factors need not be the canonical parallel-axis factors; this is why specifying the book's definition matters.

### Exercise 12.33: Square tiles marked L

**(a)** No nonidentity rotation preserves the identically upright L markings. **(b)** No mirror preserves these markings. **(c)** No nontrivial glide occurs: an orientation reversal would turn each L into a reflected letter, which no translation restores. Thus only the square-lattice translations remain. As usual for a letter-decorated pattern, the intended L is the ordinary nonsymmetric letterform in the source, not a specially designed equal-arm right-angle symbol with an extra diagonal mirror.

### Exercise 12.34: Square tiles marked E

**(a)** No nonidentity rotation preserves upright E. **(b)** Yes: reflection in a horizontal line through a row of letter centers (and suitable midway lines) preserves the pattern. Vertical or diagonal reflections do not. **(c)** No nontrivial glide in the source's sense: every glide has horizontal reflection linear part, and composing it with a horizontal mirror symmetry yields a translation symmetry. Hence it is a product of existing translation and reflection symmetries.

### Exercise 12.35: Square tiles marked H

**(a)** The angle $180°$ occurs, but $90°$ turns each H sideways and is not allowed. **(b)** Yes: horizontal and vertical mirror lines through suitable rows and columns preserve H. **(c)** No nontrivial glide: both possible reflection directions already occur as mirrors. Composing any glide with a parallel mirror symmetry gives a translation symmetry, expressing the glide as a product of symmetries of these two types.

### Exercise 12.36: The regular hexagonal tiling

**(a)** The possible angles in $(0, 180°]$ are $60°, 120°, 180°$. Rotations about hexagon centers by multiples of $60°$ preserve the tiling, and a symmetry must permute its three unoriented edge directions, excluding other angles. **(b)** Yes: lines through opposite vertices or opposite side midpoints of a hexagon are mirrors of the entire tiling. **(c)** No nontrivial glide in the book's sense. Taking a hexagon center as origin, each symmetry is a lattice translation followed by an element of the hexagon's dihedral point group; when reversing orientation, the latter is an existing reflection. Thus any glide already has the required factorization.

### Exercise 12.37: Hexagons marked with upright triangles

**(a)** The angle $120°$ occurs; $60°$ and $180°$ turn the centered upward triangle to a downward one, so they fail. **(b)** Yes: the three reflection axes of each upward equilateral triangle also preserve the hexagonal tiling. **(c)** No nontrivial glides. The remaining point group is $D_3$, and every orientation-reversing linear part is one of those existing mirrors; composing a glide with that mirror leaves a translation in the same symmetry group. This gives precisely the product excluded by the word "nontrivial."

### Exercise 12.38: Horsemen tessellation

Ignore color shading as the instructions require, but preserve the outlines and the internal horse-and-rider markings. **(a)** Half-turns ($180°$) occur; quarter-turns and smaller rotations do not, since

a horizontal horse would be turned into an absent vertical orientation. **(b)** There are no reflection axes: a mirror reverses the handed rider/horse outline and its offset adjoining shapes. **(c)** There are nontrivial glides in both horizontal and vertical directions. Reflecting and shifting by half the appropriate period matches adjacent oppositely oriented rows or columns; the reflection alone does not preserve the pattern. **(d)** Translation generators may be chosen horizontal and vertical, with directions $(1, 0)$ and $(0, 1)$. Choose the lengths as the smallest full repeats of a marked upright horse. The half-repeat displacements attached to the glides are not pure translations. These observations describe the usual *pgg* pattern.

### Exercise 12.39: Imaginary human figures tessellation

**(a)** Rotations through $90°$ and $180°$ occur. Quarter-turn centers are the junctions where four corresponding feet form a pinwheel; a quarter-turn carries one figure orientation to the next. **(b)** Yes: horizontal and vertical lines along the bilateral body axes are mirrors. There are no diagonal mirrors. Candidate axes must preserve the repeating junction pattern. In one full repeat, diagonal candidates through a four-foot junction reverse its chiral pinwheel, while shifted diagonal candidates between junction rows fail to match the neighboring head and body orientations. Horizontal and vertical mirrors can exchange different junctions, so this exclusion does not assert that every pinwheel has the same handedness. **(c)** Yes, there are nontrivial diagonal glide reflections. To see why, compose a quarter-turn symmetry with a horizontal mirror symmetry. The result reverses orientation and has the linear reflection part for a diagonal axis. Since no diagonal reflection is a symmetry, this composition must be a genuine glide reflection. It cannot be a product of a translation symmetry and a reflection symmetry: a translation does not change the linear part, so such a factorization would require an absent diagonal mirror. This distinguishes the pattern $(p4g)$ from an ordinary square grid. **(d)** Translation generators have diagonal directions $(-1, 1)$ and $(1, 1)$, with lengths taken to the first repeat of a figure in the same orientation. Following these diagonal chains, rather than moving to a differently rotated neighboring figure, identifies the translation periods. Color differences are disregarded. The printed odd-answer entry for part (c) says "No"; the diagonal glides just constructed give "Yes" under the section's own definition.

### Exercise 12.40: Reptile tessellation

**(a)** Ignoring shading, rotations through $60°, 120°, 180°$ occur. The six reptile orientations cycle around the sixfold junctions; rotations through $120°$ also occur at the threefold junctions. **(b)** No: a reflection reverses the handed arrangement of the curled bodies, feet, and tails, and that reversed arrangement is absent. **(c)** No glide reflections occur, since a glide also reverses orientation and cannot restore the handed reptile markings. **(d)** Choose two primitive translation directions along rows of identically oriented reptiles in the triangular lattice, for example $(1, 0)$ and $(1, \sqrt{3})$, with suitable lengths to reach the next full repeat. These directions have angle $60°$; the directions, rather than a scale read from the printed image, are requested. The uncolored pattern has point group $C_6$ (wallpaper type $p6$).

### Exercise 12.41: Human figures tessellation

**(a)** The possible nonzero angle at most $180°$ is $120°$, rotating the three figure orientations around their threefold junctions. A $60°$ turn would require an orientation not present. **(b)** Yes: the bilateral human figures align along mirror axes in the pattern. **(c)** No nontrivial glides in the book's sense; an existing reflection with the same linear part removes the reversing component and leaves a translation

symmetry. **(d)** Two translation-generator directions are $(0, 1)$ and $(1, \sqrt{3})$; choose their lengths to carry a figure to the next identically oriented one. These are directions only, as requested, so the lengths need not be inferred from the reproduction. Shading distinctions are not retained.

### Exercise 12.42: Cube rotations and $S_4$

**Plan.** Let a cube rotation act on the four long (unoriented) diagonals through its center. This produces a homomorphism $\Phi$ into $S_4$, because performing two rotations in succession composes their diagonal permutations. We show its kernel is trivial. If a rotation preserves all four diagonal lines, choose unit vectors $v_1, \ldots, v_4$ along them directed to the vertices of one inscribed regular tetrahedron; then $v_i \cdot v_j = -1/3$ for $i \neq j$ and $v_1 + \cdots + v_4 = 0$. The rotation sends $v_i$ to $\varepsilon_i v_i$. Preservation of the nonzero pairwise dot products gives $\varepsilon_i \varepsilon_j = 1$ for every pair, so all signs agree. If all are $+1$, three independent diagonals are fixed and the rotation is the identity. If all are $-1$, the linear map is $-I_3$, which has determinant $-1$ and is not a rotation. Thus $\Phi$ is injective. There are 24 cube rotations: choose a bottom face in six ways and a front face in four ways. Since $|S_4| = 24$, the injective map is also surjective and gives an isomorphism. It is important to use lines rather than just one endpoint of each diagonal, and to rule out central inversion as above.